
Leaderboard Incentives: Model Rankings under Strategic Post-Training

Yatong Chen^{1,2} Guanhua Zhang^{1,2} Moritz Hardt^{1,2}

Abstract

Influential benchmarks incentivize competing model providers to strategically allocate post-training resources towards improvements on the leaderboard, a phenomenon dubbed *benchmaxxing* or *training on the test task*. In this work, we initiate a principled study of the incentive structure that benchmarks induce. We model benchmarking as a Stackelberg game between a benchmark designer who chooses an evaluation protocol and multiple model providers who compete simultaneously in a subgame given by the designer’s choice. Each competitor has a model of unknown latent quality and can inflate its observed score by allocating resources to benchmark-specific improvements. First, we prove that current benchmarks generally induce games for which no Nash equilibrium between model developers exists. This result suggests one explanation for why current practice leads to misaligned incentives, prompting model providers to strategize in opaque ways. However, we prove that under mild conditions, a recently proposed evaluation protocol, called *tune-before-test*, induces a benchmark with a unique Nash equilibrium that ranks models by latent quality. This positive result demonstrates that benchmarks need not set bad incentives, even if current evaluations do.

1. Introduction

Traditionally, machine learning benchmarks came with a fixed training set, requiring that all models under comparison train on the same data. The situation has changed with large language model benchmarks that typically only provide test data, leaving the choice of training data to the model developer. This has raised the concern that model developers can inflate benchmark performance with

benchmark-specific tweaks that don’t broadly improve model capabilities. The resulting problem, called *benchmaxxing* or *training on the test task*, confounds model comparisons and may cause misleading leaderboards, as prior work shows (Dominguez-Olmedo et al., 2025; Singh et al., 2025). But the situation need not be a sign of cheating or wrongdoing. Rather, it can arise naturally when leaderboard rankings shape the returns to post-training investment. Although widely recognized, there is currently no formal understanding of the incentives that benchmarks set.

In this work, we initiate a principled study of the incentive structure that benchmarks induce. We model benchmarking as a Stackelberg game between a benchmark designer and multiple competing model developers. The designer chooses an evaluation protocol, and the model developers compete in a simultaneous-move subgame given by the designer’s choice. Each competitor has a model of a latent quality — unknown to the benchmark designer — and can inflate the observed score by allocating additional resources to benchmark-specific improvements at a cost. The designer aims to choose an evaluation protocol so that the resulting benchmark yields a ranking by latent quality at equilibrium when competitors best respond to each other.

1.1. Our Results

Our first result is descriptive and negative: Current benchmarks induce games in which generally no Nash equilibrium between model developers exists. Model developers may have persistent incentives to strategize, leading to uninterpretable leaderboards that may not reflect a ranking by latent quality. In contrast, our second result is prescriptive and positive. Under regularity conditions on post-training dynamics, the designer has a cost-effective evaluation strategy that induces a subgame with a unique Nash equilibrium that ranks models by latent quality. In addition, at this equilibrium, model developers refrain from *benchmaxxing* altogether, investing no additional effort in benchmark-specific improvements.

What makes the incentive design problem challenging is that the benchmark creator has no control over the utility each model developer has for gains on the leaderboard, does not know latent capabilities, and cannot limit how much effort model developers invest. The evaluation protocol we study tunes each model on the same small amount of task-

¹Max Planck Institute for Intelligent Systems, Tübingen

²Tübingen AI Center. Correspondence to: Yatong Chen <yatong.chen@tuebingen.mpg.de>.

specific data before evaluation. The intuition is that a small amount of task-specific preparation levels the playing field by washing out minor benchmark-specific tweaks. Previous work showed empirically that this intervention, called tune-before-test (TbT), leads to consistent model rankings across a wide range of benchmarks (Zhang et al., 2025). Our main result adds a game-theoretic justification: Tune-before-test creates incentives that lead to rankings by latent quality at equilibrium. Surprisingly, a small amount of data suffices to realize the same effect as orders of magnitude more training, as we show.

Complementing our theoretical results, we empirically examine the post-effort score assumptions in controlled case studies on representative benchmarks. In the Winogrande case study, after applying tune-before-test with 3,000 steps, the smallest additional effort required to change the ranking is 384,668 training steps. This illustrates why TbT can be effective: by moving models into a diminishing-returns regime, it raises the cost of marginal score improvements, so overtaking nearby competitors requires substantially more additional effort (see Figure 1 right).

1.2. Related Work

Benchmarks have been the key driver of machine learning progress by enabling frictionless comparison and competition (Donoho, 2024) between models. In the traditional supervised learning paradigm, benchmarks typically come with a fixed training set and a held-out test set (Garofolo et al., 1993; LeCun & Cortes, 2010; Sang & Meulder, 2003; Liberman, 2015; Hardt & Recht, 2022), enabling relatively controlled comparisons across methods. Although test-set reuse (Duda & Hart, 1974) can erode classical statistical guarantees (Dwork et al., 2015a;b; Blum & Hardt, 2015; Mania et al., 2019), fixed train/test splits have historically supported relatively robust model comparison and have helped establish widely adopted baselines, exemplified by the ImageNet challenge (Deng et al., 2009; Russakovsky et al., 2014) and its role in accelerating the adoption of deep learning (Krizhevsky et al., 2012; Goodfellow et al., 2016; He et al., 2015).

A key reason benchmarks are useful in practice is *external validity* (Liao, 2021; Salaudeen et al., 2025): performance and rankings on one benchmark often correlate with performance on related datasets and tasks (Yadav & Bottou, 2019; Recht et al., 2019; Miller et al., 2020a), allowing practitioners to select strong models with some confidence. For instance, studies have shown that ImageNet rankings transfer well to other image datasets (Kornblith et al., 2019; Salaudeen & Hardt, 2024). In the large language model (LLM) era, however, many influential benchmarks provide primarily test instances and a scoring protocol (Patwardhan et al., 2025; Glazer et al., 2024; Zhou et al., 2023a; Jain et al., 2024), leaving training data and post-training

choices largely unconstrained (Raffel et al., 2019; Albalak et al., 2024; Guha et al., 2025; Li et al., 2024). This change expands the space of strategic choices available to model providers and can lead to substantial ranking variation across benchmarks (Huan et al., 2025; Zhang & Hardt, 2024; Liang et al., 2023; Fourrier et al., 2024; Hardt, 2025), even among benchmarks that aim to measure similar capabilities.

One mechanism behind the ranking variation is what Dominguez-Olmedo et al. (2025) call *training on the test task*. Model providers conduct benchmark-aware post-training (Touvron et al., 2023) and data curation (Guha et al., 2025) to achieve large gains on measured tasks without comparable improvements in general capability (Zhou et al., 2023b; Singh et al., 2025). Training on the test task differs from training on the test set (Yang et al., 2023), or data contamination (Jiang et al., 2024; Yang et al., 2023; Bordt et al., 2024), where models train directly on the test data. Instead, it captures optimization to the evaluation’s task distribution or protocol, such as curating instruction data that matches a benchmark’s format and rubrics or tuning with feedback aligned to its scoring procedure (Dominguez-Olmedo et al., 2025). Because such task-level alignment is difficult to detect and is often not explicitly ruled out by benchmark rules, it can confound leaderboard interpretation: scores conflate latent capability (Ruan et al., 2024) with benchmark-specific effort (Schaeffer et al., 2023). Model providers can strategically choose how much benchmark-specific effort to invest per benchmark, which yields inconsistent rankings.

A natural way to mitigate benchmark-specific advantages is to reduce heterogeneity in how models are prepared for the evaluation. Zhang et al. (2025) propose tune-before-test (TbT), in which all models are fine-tuned on benchmark-specific data before evaluation to equalize preparation. Their empirical results show that TbT can restore ranking consistency across tasks from different domains and reveal that post-TbT scores are dominated by a low-dimensional latent capability factor. While Zhang et al. (2025) motivate TbT as a post-hoc correction for observed rankings, we study it as an ex ante mechanism design choice (Manheim & Garrabrant, 2018). In our setting, a benchmark designer commits to an evaluation protocol (including a TbT baseline), anticipating that competing model providers will respond strategically by allocating additional post-training effort. We ask when an equilibrium exists in this competition and whether equilibrium rankings recover a ranking by latent capability. Treating TbT as a designer-controlled policy parameter instead allows us to isolate its incentive effects and ask how much evaluator-side adaptation is required to stabilize rankings.

Relevant to our work is the literature on strategic classification (Brückner & Scheffer, 2011; Hardt et al., 2016), which studies decision-making when individuals may adapt their

features in response to a deployed model. See Rosenfeld (2024) for a survey. These interactions are typically modeled as a Stackelberg game, where a decision-maker commits to a classifier and strategic individuals best respond myopically to the classifier. An important question in this literature is how to incentivize genuine improvements rather than superficial *gaming* (Kleinberg & Raghavan, 2020; Miller et al., 2020b; Alon et al., 2020; Chen et al., 2023). Our work shares this focus on incentives. In a departure from the classification setting, however, ranking inherently creates competition between participants: when one model moves up in rank, another must move down. The leaderboard therefore induces a game between competitors. This aspect of our work also connects to recent work on markets induced by predictive systems (Einav & Rosenfeld, 2025; Sommer et al., 2025). Closely related is the work on *strategic ranking* (Liu et al., 2022), where a continuous population of individuals competes for acceptance by an institution, such as a school or an employer. The institution may set the rewards for achieving a given quantile rank within the continuous population. The results characterize how quantile-based incentive design shapes effort, welfare, and fairness at equilibrium. Our work shares this focus on rank-based competition, but studies the setting of machine learning benchmarking. Here, the designer cannot directly control the downstream rewards associated with a higher rank, such as more attention or increased reputation, since these are exogenous. Instead, the designer’s primary lever is not the reward design but the evaluation protocol itself, which needs to make sense for a machine learning benchmark. Second, Liu et al. (2022) model a continuum of agents, leading to a smooth equilibrium characterization with monotone effort profiles. In contrast, leaderboard competition typically involves a finite number of strategic model providers, where small performance changes can swap adjacent ranks and generate discontinuous payoff changes. These local “just-overtake” incentives can, in our setting, make equilibrium stability itself a central design concern. Finally, strategic ranking also relates to a vast literature in economics on designing tournaments and contests, see (Corchón, 2007; Connelly et al., 2014; Fu & Wu, 2019), that we cannot survey here.

2. Preliminaries

Capability, effort, and post-effort score. Consider a set of LLM model developers $N = \{1, \dots, n\}$ who submit a model to a benchmark leaderboard curated by a leaderboard designer. Each model has a latent variable $\theta_i \in \mathbb{R}_{\geq 0}$ capturing general model capabilities. This abstraction is empirically motivated by evidence that performance across diverse language-model benchmarks admits a low-dimensional structure and is often dominated by a general capability factor (Ruan et al., 2024; Zhang et al., 2025). The setting extends straightforwardly to a multi-dimensional latent capabilities vector $\vec{\theta}_i \in \mathbb{R}_{\geq 0}^d$ with a benchmark-specific

coefficient vector $\mathbf{w}$ so that $\theta_i = \langle \mathbf{w}, \vec{\theta}_i \rangle$. Without loss of generality, we index models in decreasing order of latent capability, i.e., $\theta_1 > \theta_2 > \dots > \theta_n$. Capabilities are known to model developers but not to leaderboard designer.

Before evaluation, a model may undergo additional *benchmark-specific adaptation*. We represent the *total* amount of benchmark-specific preparation by a scalar $e \in \mathbb{R}_{\geq 0}$, which may be performed by the model developer, and/or applied uniformly by the leaderboard designer (as in tune-before-test). Following the empirical tune-before-test approach of (Zhang et al., 2025), we use the amount of benchmark-specific fine-tuning data (e.g., number of training examples) as a concrete and measurable proxy for this adaptation. We model benchmark performance through a *post-effort score* mapping $v = v(\theta, e)$:

Definition 2.1 (Post-Effort Score). The *post-effort score* of a model with capability θ and total benchmark-specific training effort e is $v(\theta, e) : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \rightarrow [0, 1]$.

The function $v(\theta, e)$ captures how capability and benchmark-specific effort translate into its actual benchmark performance. For simplicity, we model v deterministically; equivalently, our results can be interpreted in terms of expected scores under evaluation noise.

Leaderboard rewards. Let $\{R_j\}_{j=1}^n$ denote the reward assigned to the model ranked j , where rewards are non-increasing in rank, i.e., $R_1 \geq R_2 \geq \dots \geq R_n \geq 0$. This means higher-ranked models receive greater downstream benefits (e.g., more downstream users, higher visibility) from their leaderboard positions.

Example 2.2. Two canonical leaderboard reward settings:

1. *Winner-take-all*: only the top-ranked model receives a reward, i.e., $R_1 > 0$ and $R_j = 0$ for all $j \neq 1$.
2. *Top- k rewards*: the top k models receive the same reward, i.e., $R_j = R > 0$ for $j \in [k]$ and $R_j = 0$ for $j > k$.

Tune-before-test methodology. We model tune-before-test (TbT) as a designer-chosen baseline effort $\Delta^{ibt} \geq 0$ that applies the same amount of benchmark-specific fine-tuning to every submitted model before evaluation. We treat TbT as an explicit part of the evaluation protocol: the leaderboard designer commits to Δ^{ibt} , and model developers choose any additional benchmark-specific effort in response. Setting $\Delta^{ibt} = 0$ recovers the standard evaluation without TbT, so the conventional leaderboard protocol is a special case of our framework.

3. Stackelberg Ranking Game

We model leaderboard evaluation as a Stackelberg game with a single leader and multiple followers (Von Stengel &

Zamir, 2010). The leader (the leaderboard designer) first commits to an evaluation protocol, after which the followers (model developers) simultaneously choose benchmark-specific effort. We study outcomes in which the developers play a Nash equilibrium of the induced follower game, and the leader chooses its action, anticipating this equilibrium response. Such an outcome is often referred to as a *Stackelberg–Nash equilibrium* (Marchesi, 2021).

Definition 3.1 (Stackelberg Ranking Game). The players are a leaderboard designer (the leader) and a set of model developers $N = \{1, \dots, n\}$ (the followers). Model developers have latent capabilities $(\theta_i)_{i=1}^n$, which are common knowledge among the model developers but unknown to the designer. A public rank-based reward $\{R_j\}_{j=1}^n$ assigns reward R_j to the model ranked j . The sequence of actions is as follows:

- The designer commits to a benchmark and a tune-before-test baseline Δ^{tb} ≥ 0 .
- After observing Δ^{tb} , developers simultaneously choose additional effort levels $e_i \geq 0$.

Scores are realized as $v_i = v(\theta_i, e_i + \Delta^{tb})$, and models are ranked by $\mathbf{v} = (v_1, \dots, v_n)$ (ties broken deterministically). Both parties’ utilities are specified in Definition 3.2 and Definition 3.4.

3.1. Model Developer’s Utility and Externalities

Model developers are modeled as rational agents whose utility equals the rank-based reward minus the cost of additional effort. We use a cost function $c : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ to capture the effort costs¹. Because rewards depend on rank, incentives are inherently interdependent: improving one model’s rank necessarily lowers another’s. As a result, a developer’s optimal effort depends not only on its own capability and cost, but also on competitors’ efforts. Let $\mathbf{e}_{-i}$ denote the vector of the efforts of all models except i . We define the utility of developer i as follows:

Definition 3.2 (Model Developer’s Utility). Given a rank-based reward $(R_j)_{j=1}^n$ and a fixed tune-before-test adjustment level $\Delta^{tb} \geq 0$, the utility of model i with capability θ_i and effort e_i is

$$U_i(e_i; \mathbf{e}_{-i}, \Delta^{tb}) = R_{\text{rank}(v_i)} - c(e_i), \quad (1)$$

$$v_i = v(\theta_i, \Delta^{tb} + e_i), \text{rank}(v_i) := 1 + \sum_{j \in N \setminus \{i\}} \mathbb{I}\{v_j > v_i\}.$$

¹We assume a common cost function for all model developers for simplicity. In Proposition B.1, we show that *multiplicatively separable* costs $(C_i(e_i; \theta_i) = \gamma_i c(e_i))$ yield an equivalent follower game after rescaling rewards.

Nash equilibrium of the induced follower game. For a fixed TbT level Δ^{tb} , the developers’ interaction forms the induced follower game, for which we use pure-strategy Nash equilibrium (Nash, 1950) as the solution concept. Intuitively, an effort profile is at equilibrium if no developer can improve its utility by unilaterally changing the amount of benchmark-specific effort it invests:

Definition 3.3 (Follower Game’s Nash Equilibrium). Fix a TbT level Δ^{tb} . An effort profile $\mathbf{e}^* = (e_1^*, \dots, e_n^*)$ is a *pure-strategy Nash equilibrium* (PNE) of the induced follower game if, for every model developer $i \in N$,

$$e_i^* \in \arg \max_{e \geq 0} U_i(e; \mathbf{e}_{-i}^*, \Delta^{tb}),$$

where $\mathbf{e}_{-i}^*$ denotes the equilibrium efforts of other models.

Although real-world leaderboard competition involves repeated submissions over time, equilibrium analysis still provides a useful baseline: it clarifies whether incentives can ever settle, or whether the leaderboard instead induces persistent “arms-race” behavior.

3.2. Leaderboard Designer’s Utility

The leaderboard designer evaluates a benchmark by whether its induced ranking reflects latent capabilities, while paying for interventions such as TbT. A simple way to capture this objective is through a *ranking correctness* criterion:

Definition 3.4. (Leaderboard Designer’s Utility) Given a tune-before-test level Δ^{tb} and follower effort profile $\mathbf{e} = (e_1, \dots, e_n)$ inducing scores $v_i = v(\theta_i, e_i + \Delta^{tb})$, the designer’s utility is

$$U^L(\Delta^{tb}; \mathbf{e}) = R^L \mathbb{I}\{\mathcal{E}_{\text{rank}}\} - n c^L(\Delta^{tb}), \quad (2)$$

where $\mathcal{E}_{\text{rank}}$ denotes the event that $\text{Rank}(v_i) = \text{Rank}(\theta_i)$ for all $i \in [n]$, $R^L \gg 0$ is the reward for a capability-consistent ranking, and $c^L(\Delta^{tb})$ is the per-model cost of applying TbT.

This binary objective captures the designer’s primary concern: whether the leaderboard correctly orders models by latent capability. Alternatives could use smoother rank-correlation metrics such as Kendall’s τ (Kendall, 1938). We assume R^L is large enough that any capability-consistent ranking is preferred to any inconsistent one, even after TbT costs, and that $c^L(\Delta^{tb})$ is increasing. Thus, among TbT levels that induce a capability-preserving follower equilibrium, the designer prefers the cheapest one, yielding the Stackelberg–Nash equilibrium defined below.

3.3. Stackelberg–Nash Equilibrium

Formally, a pair $(\Delta^{tb*}, \mathbf{e}^*)$ is a *Stackelberg–Nash equilibrium* if: (i) $\mathbf{e}^*$ is a pure-strategy Nash equilibrium of the induced follower game under Δ^{tb*} , and (ii) Δ^{tb*} maximizes the designer’s utility anticipating this response.

We begin by fixing Δ^{tbT} and analyzing the induced follower game. Understanding the existence of follower equilibria and the rankings they generate is a prerequisite for determining the designer’s optimal policy. In Section 5, we then study how to choose the optimal TbT level Δ^{tbT*} to maximize the leaderboard designer’s utility.

4. Equilibrium Analysis of the Follower Game

What incentives does a rank-based leaderboard create for strategic model developers? When model developers can invest in additional benchmark-specific post-training to improve performance, a central question is whether the induced competition admits a pure-strategy Nash equilibrium, that is, whether incentives can settle at a stable effort profile.

Perhaps surprisingly, our first result is partially optimistic: whenever a pure-strategy equilibrium of the induced follower subgame exists, the resulting leaderboard ranking must preserve the latent capability ordering (Proposition 4.3). In this case, any strategic post-training should still lead to a stable and correct ranking. However, as we point out later, the more subtle issue is that such equilibria need not exist. We show that equilibrium existence depends critically on the *reward gaps* between adjacent ranks (Theorem 4.6). When rewards are sufficiently large, developers face persistent incentives to “just overtake” nearby competitors, leading to arms-race dynamics.

4.1. Assumptions and Empirical Verifications

To analyze equilibrium behavior in the induced follower game, we need to make some structural assumptions on (i) the cost of benchmark-specific post-training, and (ii) how benchmark performance depends on a model’s capability and post-training effort.

Initial benchmark-specific data curation may be inexpensive, but getting more data typically becomes progressively more costly (DatologyAI, 2024). To capture this, we assume that the cost of effort is non-decreasing and convex:

Assumption 4.1 (Cost function c). The cost of benchmark-specific post-training $c : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is non-decreasing and convex, with $c(0) = 0$ and $\lim_{e \rightarrow \infty} c(e) = \infty$.

Next, we impose structure on the post-effort score function:

Assumption 4.2 (Post-effort score function v). Let $v : \Theta \times \mathbb{R}_{\geq 0} \rightarrow [0, 1]$ denote the post-effort score function, where $\theta \in \Theta \subseteq \mathbb{R}_{\geq 0}$ is the capability and $e \in \mathbb{R}_{\geq 0}$ is the total effort. Assume v is continuous and that for all (θ, e) :

C1. (Monotonicity in capability) Holding effort fixed, higher capability yields a higher score: $\partial_{\theta} v(\theta, e) > 0$.

C2. (Diminishing returns and saturation in effort) Effort weakly improves performance but with diminishing marginal returns, and scores converge to a finite limit: $\partial_e v(\theta, e) \geq 0$, $\partial_{ee} v(\theta, e) \leq 0$, and $v^{\infty}(\theta) := \lim_{e \rightarrow \infty} v(\theta, e)$ exists and is finite.

C3. (Non-decreasing effort gaps) Let $e^{\text{req}}(s; \theta) := \inf\{e \geq 0 : v(\theta, e) \geq s\}$ denote the minimal effort required for capability θ to reach target score $s \in [0, 1]$. For any $\theta' > \theta$, the effort advantage of higher capability, $e^{\text{req}}(s; \theta) - e^{\text{req}}(s; \theta')$, is (weakly) nondecreasing in s .

These conditions capture three widely observed regularities in post-training scaling behavior: higher-capability models perform better at any fixed effort (C1), benchmark-specific training exhibits diminishing returns and saturation as models approach the benchmark’s performance ceiling (C2). C3 further requires that as the target score increases, it does not become easier for a lower-capability model to close the gap through post-training alone. This is a standard single-crossing-type regularity condition in economic theory (Topkis, 1998), and is consistent with empirical observations that stronger foundation models tend to make more effective use of additional training, especially at a higher target score (Wei et al., 2021).

Example: generalized power-law scaling. As an illustrative functional form consistent with Assumption 4.2, consider a generalized scaling law motivated by empirical studies of post-training behavior (Ruan et al., 2024; Finnveden, 2020; Owen, 2024). Let

$$\tilde{v}(\theta, e) := \frac{v(\theta, e) - L(\theta)}{U(\theta) - L(\theta)} \in [0, 1]$$

denote a normalized score, where $L(\theta)$ and $U(\theta)$ represent model-specific lower and upper performance levels. Suppose

$$\sigma^{-1}(\tilde{v}(\theta, e)) = \alpha(\theta) + \beta(\theta) \log(1 + e), \quad (3)$$

where σ^{-1} is the logit link function, $\alpha(\theta)$ is the baseline performance on the logit scale, and $\beta(\theta) > 0$ is the scaling coefficient governing how efficiently extra compute improves performance. If $\alpha(\theta)$ and $\beta(\theta)$ are both weakly increasing in θ , this specification satisfies all conditions in Assumption 4.2 (C1–C3).

We also provide empirical evidence for Assumption 4.2 using controlled post-training experiments within a single model family. Because latent capability θ and prior benchmark-specific post-training effort are unobserved, we restrict attention to *Qwen2.5* models (Yang et al., 2024). Within this family, we use model size as a proxy for capability θ , and we treat the models’ pre-existing post-training as approximately the same across sizes. We then apply additional benchmark-specific post-training to each model on

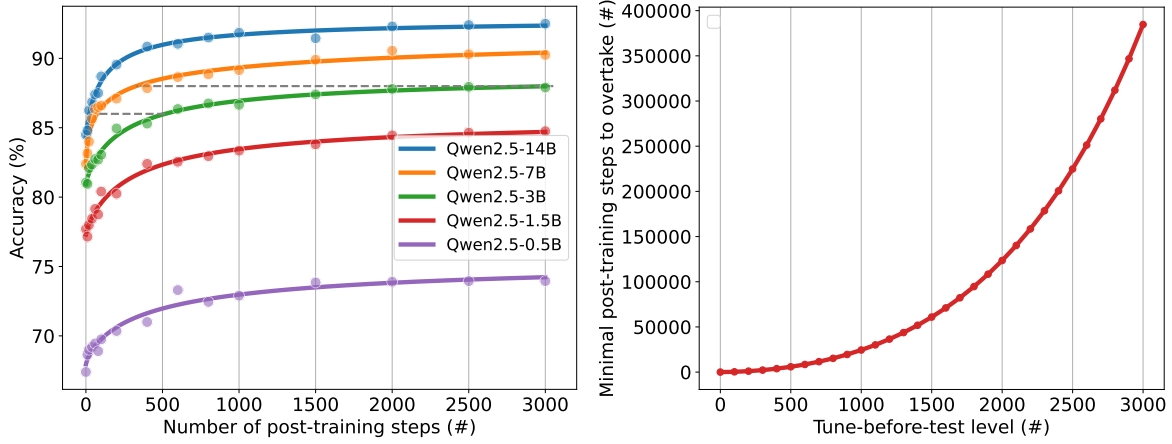


Figure 1. Left: Continued post-training trajectories of *Qwen2.5* models of different sizes on *Winogrande*. Here, we use model size as a proxy for the model’s latent capability θ . The x -axis denotes the amount of post-training steps (each step corresponds to 8 data points), reflecting post-training effort e . The y -axis denotes accuracy on the validation set, i.e., $v(\theta, e)$. For each model, we fit a curve following Equation (3). The empirical results align with the assumptions of *monotonicity in capability*, *diminishing returns and saturation in effort*, and *non-decreasing effort gaps* in Assumption 4.2. See Appendix E for additional details and results for the other eight benchmarks. **Right:** For each tune-before-test level Δ^{ibt} (the amount of benchmark-specific finetuning steps, x -axis), we calculate the minimal additional steps required (y -axis) to change the ranking for at least one model, i.e., $\min_{r \in \{2, \dots, n\}} e_r^{\text{req}}(\Delta^{ibt})$, based on the fitted curves on the left. With $\Delta^{ibt} = 3,000$, at least 384,668 training steps are needed to change the ranking of one model.

the benchmark’s training set, plotting performance on the validation set as a function of added effort e (measured by the incremental number of training steps).

Figure 1 (left) shows the results on *Winogrande*, a large-scale commonsense pronoun resolution benchmark introduced by Sakaguchi et al., while results for the other eight benchmarks are in Appendix E. For each model size, we fit the generalized power-law specification in Equation (3). The fitted curves track the observed points closely and are consistent with Assumption 4.2. First, at any fixed e , larger models achieve better performance (monotonicity in θ , C1). Second, gains from additional effort diminish and scores approach a plateau (concavity and saturation in e , C2). Third, the horizontal distance between curves—interpretable as the extra effort required for a smaller model to match a larger model’s target score—does not shrink at higher target accuracies (increasing effort gaps, C3). For instance, the implied effort gap between *Qwen2.5-3B* and *Qwen2.5-7B* is larger at 88% accuracy than at 86%, as illustrated by the dashed guides in Figure 1 (left).

To assess whether these patterns generalize beyond a single model family and a fixed fine-tuning recipe, we extend the experiments to a cross-family set of *Qwen2.5*, *Llama 3*, and *Gemma 2* models, and sweep the fine-tuning learning rate in both within- and cross-family settings. In both cases, both the qualitative trajectories and the rankings they induce remain stable, with higher-capability models eventually overtaking those that had a head start. See Appendices E.1 and E.2 for details.

4.2. Order Properties of Equilibrium Profiles

Our first result shows that if the induced follower game admits a pure-strategy equilibrium $\mathbf{e}^*$, the resulting leaderboard ordering must respect the latent capability ordering (Proposition 4.3). Intuitively, this means that if all developers choose benchmark-specific post-training optimally, strategic fine-tuning alone cannot cause a lower-capability model to strictly out-rank a higher-capability one:

Proposition 4.3. *Under Assumption 4.1 and Assumption 4.2, fix any tune-before-test adjustment level $\Delta^{ibt} \geq 0$. If the follower game admits a pure-Nash equilibrium $\mathbf{e}^*$, then for any i, j ,*

$$\theta_i > \theta_j \Rightarrow v(\theta_i, \Delta^{ibt} + e_i^*) \geq v(\theta_j, \Delta^{ibt} + e_j^*).$$

In particular, post-effort scores at equilibrium preserve the latent capability ordering up to ties.

Proposition 4.3 offers reassuring news: if the leaderboard competition ever settles, it will settle in the *right* order. However, this positive result comes with an important caveat. As we show next, such equilibria need not exist at all. In particular, if the reward gap between some adjacent ranks is large relative to the cost of overtaking, namely, if there exists $r \in \{2, \dots, n\}$ with $R_{r-1} - R_r$ sufficiently large, the induced follower game fails to admit any pure-strategy Nash equilibrium. In this regime, developers face persistent incentives to “just overtake” nearby competitors, leading to arms-race dynamics rather than convergence.

4.3. Existence of Pure-Nash Equilibrium

Fixing a TbT Δ^{ibt} , we analyze the induced follower game among developers and characterize when it admits a pure-strategy Nash equilibrium, and when it does not.

All-Zero additional effort profile. A natural reference point is the all-zero additional-effort profile $\mathbf{e} = (0, \dots, 0)$, where model i attains the baseline post-TbT score

$$s_i(\Delta^{ibt}) = v(\theta_i, \Delta^{ibt}).$$

By monotonicity in capability (C1), we have $s_1(\Delta^{ibt}) > \dots > s_n(\Delta^{ibt})$. Because rewards depend only on rank and effort costs are nondecreasing, any profitable deviation must be an *overtaking move*, and the cheapest such deviation is to overtake the adjacent competitor directly above:

Definition 4.4 (Just-Overtake Effort at TbT Level Δ^{ibt}). Fix a TbT level $\Delta^{ibt} \geq 0$. For any rank $r \in \{2, \dots, n\}$, define the “just-overtake” effort as

$$e_r^{\text{req}}(\Delta^{ibt}) := \inf \left\{ e \geq 0 : v(\theta_r, \Delta^{ibt} + e) > s_{r-1}(\Delta^{ibt}) \right\}.$$

Equivalently, $e_r^{\text{req}}(\Delta^{ibt})$ measures how difficult it is for the model designer r to “climb” from rank r to rank $r - 1$ starting from any common TbT baseline. When $e_r^{\text{req}}(\Delta^{ibt})$ is small, even minor fine-tuning can change rank, creating strong overtaking incentives. When it is large, small investments are unlikely to affect the leaderboard ordering. The following proposition precisely states when the all-zero effort profile is a pure Nash equilibrium strategy of the induced follower game:

Proposition 4.5 (Zero-effort equilibrium condition). Fix any $\Delta^{ibt} \geq 0$. The all-zero profile $\mathbf{e} = (0, \dots, 0)$ is a PNE if and only if, for every $r \in \{2, \dots, n\}$,

$$c(e_r^{\text{req}}(\Delta^{ibt})) \geq R_{r-1} - R_r. \quad (4)$$

In words, with TbT level Δ^{ibt} , no agent can profitably “just overtake” the model immediately above it: the minimal cost required to surpass the adjacent baseline score exceeds the corresponding reward gap.

Proposition 4.5 further implies that if the adjacent “just-overtake” condition fails for some model designer, then no pure-strategy equilibrium exists:

Theorem 4.6 (Equilibrium structure and nonexistence). Fix any $\Delta^{ibt} \geq 0$, and suppose ties are broken deterministically in a way that does not favor the lower-capability model². Then any PNE, if it exists, must be the all-zero profile

²Under random tie-breaking, a lower-capability model may benefit from exerting positive effort to enter a tie lottery, so the conclusion that every PNE is all-zero need not hold without further qualification. This is mainly a technical issue: exact ties are rare in practice because leaderboard scores are usually reported at high precision or resolved by fixed secondary rules.

$\mathbf{e} = (0, \dots, 0)$. Moreover, if there exists $r \in \{2, \dots, n\}$ such that

$$c(e_r^{\text{req}}(\Delta^{ibt})) < R_{r-1} - R_r, \quad (5)$$

then the induced follower game admits no PNE.

Proof sketch. Suppose a PNE $\mathbf{e}^*$ exists. By Proposition 4.3, post-effort scores are weakly ordered by capability at equilibrium. Since ties are broken deterministically against the lower-capability model, model n remains last-ranked whenever it only ties a higher-capability model. Thus, it cannot gain from positive effort that does not strictly overtake model $n - 1$, and it weakly improves by setting $e_n = 0$. Hence $e_n^* = 0$. Given $e_n^* = 0$, model $n - 1$ strictly outranks model n at zero effort by (C1) and cannot lose rank by reducing effort, so $e_{n-1}^* = 0$. Repeating this argument inductively yields $e_i^* = 0$ for all i . Therefore, any PNE must be the all-zero profile $\mathbf{e} = (0, \dots, 0)$.

If $c(e_r^{\text{req}}(\Delta^{ibt})) < R_{r-1} - R_r$ for some $r \in \{2, \dots, n\}$, then at $\mathbf{e} = (0, \dots, 0)$ model r can profitably deviate by exerting effort $e_r^{\text{req}}(\Delta^{ibt})$, thereby overtaking model $r - 1$ and increasing its reward by $R_{r-1} - R_r$ at lower cost. Hence $\mathbf{e} = (0, \dots, 0)$ is not a PNE. Since any PNE must be all-zero, no PNE exists. $\square$

The theorem shows that incentives are governed locally by the adjacent reward gap $R_{r-1} - R_r$: moving from rank r to rank $r - 1$ is profitable only if this gain is large enough relative to the corresponding overtaking cost. This gives a simple interpretation of common reward schemes. In winner-take-all schemes, incentives are concentrated at the top rank; in top-k schemes, they are concentrated near the cutoff into the rewarded set. By contrast, under a smoothly decaying reward scheme, adjacent reward gaps are spread more evenly across ranks and may all be small. In that case, no single rank improvement creates a large reward jump, so overtaking incentives are correspondingly weaker throughout the leaderboard.

Our analysis also suggests that the main failure mode of leaderboards is not that they converge to a *stable but incorrect* ranking. Instead, leaderboards can fail because they may incentivize *continuous competitive fine-tuning*: developers are motivated to repeatedly invest in benchmark-specific optimization simply to gain (or defend) a small rank advantage (Equation (5)).

This regime is particularly relevant when benchmarks provide limited separation between models, so that even small performance gains translate into meaningful rank changes. Such situations can arise on *saturated* benchmarks where frontier models perform similarly (e.g., MMLU (Hendrycks et al., 2020) or HellaSwag (Zellers et al., 2019b)), as well as on *very difficult or early-stage* benchmarks where all

models perform poorly and remain tightly clustered (e.g., Humanity’s Last Exam (Phan et al., 2025)).

5. Tune-before-Test Aligns Benchmark Incentives

We now turn to the leaderboard designer’s perspective and ask: if large reward gaps can preclude equilibrium in the induced follower game, how should the designer choose Δ^{ibt} to stabilize incentives? At a high level, tune-before-test shifts the leaderboard’s “operating point” by applying the same amount of benchmark-specific training to all models, so the baseline scores become $s_i(\Delta^{ibt}) = v(\theta_i, \Delta^{ibt})$. This has two conceptually distinct effects.

By moving models closer to their benchmark-specific performance limits, TbT *directly* reduces the leaderboard’s sensitivity to who prepared more aggressively from scratch. In the limit as models approach their saturation levels $v^\infty(\theta) = \lim_{e \rightarrow \infty} v(\theta, e)$, rankings depend only on capability. Under monotonicity in capability (C1), saturation yields a capability-consistent ordering:

Proposition 5.1 (Rank preservation at saturation). *Suppose the post-effort score $v(\theta, e)$ satisfies $\partial_\theta v(\theta, e) > 0$ (C1). Then the saturated score $v^\infty(\theta) := \lim_{e \rightarrow \infty} v(\theta, e)$ is increasing in θ .*

More importantly, TbT also weakens the incentives for strategic post-training. Since $v(\theta, e)$ exhibits diminishing returns in effort (C2), increasing Δ^{ibt} pushes all models into a regime where further improvements are harder.

5.1. TbT Increases Leaderboard Climbing Cost

To formalize this incentive effect, recall $e_r^{\text{req}}(\Delta^{ibt})$, the minimal additional effort required for model r to overtake the adjacent competitor at baseline Δ^{ibt} (Definition 4.4). This precisely captures the marginal difficulty of climbing the leaderboard. The next lemma shows that increasing the TbT baseline monotonically raises this overtaking cost:

Lemma 5.2 (TbT Increases the Leaderboard Climbing Cost). *For any $\Delta^{ibt} \geq 0$ and $r \in \{2, \dots, n\}$, the minimal just-overtake effort $e_r^{\text{req}}(\Delta^{ibt})$, and thus also the corresponding cost $c(e_r^{\text{req}}(\Delta^{ibt}))$, is non-decreasing in Δ^{ibt} .*

Figure 1 (right) plots the minimal just-overtake effort across all models $\min_{r \in \{2, \dots, n\}} e_r^{\text{req}}(\Delta^{ibt})$ (y -axis), as a function of Δ^{ibt} (x -axis). We estimate $e_r^{\text{req}}(\Delta^{ibt})$ using the fitted post-training trajectories shown in the left panel. For example, for model Qwen2.5-7B at $\Delta^{ibt} = 500$, we first compute the accuracy of Qwen2.5-14B at $\Delta^{ibt} = 500$, 91.0%. We then calculate how much additional training data Qwen2.5-7B requires to reach 91.0% based on the fitted curve, which is 5,890. We repeat this for all models (except for the best model Qwen2.5-14B) and plot the minimum across all rank $r \geq 2$. The resulting curve shows that leaderboard climbing

becomes rapidly more difficult as Δ^{ibt} grows. At $\Delta^{ibt} = 0$, only 18 additional steps are needed to change the ranking, whereas at $\Delta^{ibt} = 3,000$, the required effort rises to 384,668 steps.

5.2. TbT Restores Equilibrium Existence

Since increasing TbT raises the overtaking effort required, the all-zero equilibrium condition in Proposition 4.5 becomes easier to satisfy at higher TbT levels:

Proposition 5.3. *Given two tune-before-test adjustment levels $0 \leq \Delta_1^{ibt} \leq \Delta_2^{ibt}$, if the all-zero effort profile $\mathbf{e} = (0, \dots, 0)$ is a PNE under Δ_1^{ibt} , it is also a PNE under Δ_2^{ibt} .*

Thus, TbT has a monotone stabilizing effect on post-training incentives: once no model developer finds it profitable to exert additional effort, this remains true under any larger TbT intervention.

5.3. How Much TbT is Enough?

In practice, a benchmark evaluator may not have enough compute to apply a large TbT adjustment to every submitted model, due to resource or cost constraints. This raises a natural question: what is the smallest TbT level that eliminates incentives for further strategic fine-tuning? This motivates us to view TbT as an incentive-control parameter with a corresponding threshold.

Definition 5.4 (Stabilizing TbT Threshold). Define the *stabilizing TbT threshold* as

$$\Delta^{ibt*} := \inf \left\{ \Delta^{ibt} \geq 0 : c(e_r^{\text{req}}(\Delta^{ibt})) \geq R_{r-1} - R_r, \right. \\ \left. \forall r \in \{2, \dots, n\} \right\}. \quad (6)$$

that is, the smallest TbT adjustment for which no model can profitably overtake the model directly above it at the baseline level.

At Δ^{ibt*} , the cost of overtaking the immediate neighbor exceeds the corresponding reward gain for every adjacent pair. By Proposition 5.3, once this condition holds, it continues to hold for all larger TbT levels. Hence, any $\Delta^{ibt} \geq \Delta^{ibt*}$ eliminates profitable overtaking incentives:

Corollary 5.5. *For any $\Delta^{ibt} \geq \Delta^{ibt*}$, the induced follower game admits a PNE in which all model developers choose zero additional effort, i.e., $\mathbf{e}^* = (0, \dots, 0)$. In particular, no model developer engages in strategic post-training.*

Under the leaderboard designer objective in Definition 3.4, Δ^{ibt*} is also the optimal TbT choice: since it is the smallest intervention that induces a follower equilibrium preserving the true capability ordering, it therefore achieves the designer’s ranking objective at minimum cost.

5.4. Case Study Under Generalized Power-Law

To build intuition for the scale of the stabilizing TbT level Δ^{ibt*} , consider the generalized scaling law

$$\sigma^{-1}(\tilde{v}(\theta, e)) = \alpha(\theta) + \beta(\theta) \log(1 + e),$$

with costs satisfying $c(e) \geq \kappa e$ for some $\kappa > 0$. For an adjacent pair $(r-1, r)$, let $\rho_r := (R_{r-1} - R_r)/\kappa$ denote the reward gap in units of effort, and let $e_r^{\text{req}}(0)$ be the baseline catch-up effort without TbT. Define

$$\lambda_r := \frac{\rho_r}{e_r^{\text{req}}(0)}, \quad \gamma_r := \frac{\beta(\theta_r)}{\beta(\theta_{r-1})} \leq 1.$$

Here, λ_r measures the effective overtaking incentive, while γ_r captures the relative learning rate between two models.

Proposition 5.6 (Stabilizing TbT under generalized scaling). *Under the generalized scaling law, and if $c(e) \geq \kappa e$, there exists a stabilizing TbT threshold Δ^{ibt*}_r such that any $\Delta^{ibt} \geq \Delta^{ibt*}_r$ eliminates profitable overtaking deviations from rank r to rank $r-1$. Moreover, in the regime where catch-up remains feasible, $\Delta^{ibt*}_r = O(\lambda_r^{\gamma_r})$, up to a constant factor depending only on $e_r^{\text{req}}(0)$.*

The exponent γ_r determines how quickly the required TbT level grows with the overtaking incentive. When γ_r is small, the stronger model benefits more from effort, so rankings are easier to stabilize; when $\gamma_r \approx 1$, the two models benefit similarly, and a larger TbT baseline is needed. Thus, the stabilizing threshold grows polynomially in λ_r , but more slowly when effort-response gaps are larger.

To stabilize the full leaderboard, it suffices to choose

$$\Delta^{ibt*} := \max_{r \in \{2, \dots, n\}} \Delta^{ibt*}_r,$$

namely, the global threshold is determined by the hardest adjacent pair.

A concrete example. We estimate the baseline catch-up difficulty $e_r^{\text{req}}(0)$ and the learning-rate ratio γ_r from the fitted trajectories in Figure 1. For the hardest-to-stabilize pair, we estimate $e_r^{\text{req}}(0) \approx 13.7$ and $\gamma_r \approx 0.33$. Suppose the corresponding reward gap translates to $\rho_r = 1000$ units of effort, so the effective incentive is $\lambda_r = \frac{\rho_r}{e_r^{\text{req}}(0)} \approx \frac{1000}{13.7} \approx 73$. Using Proposition 5.6, a stabilizing TbT level is

$$\Delta_r^{ibt*} \approx \left(\frac{\rho_r}{e_r^{\text{req}}(0)} \right)^{\gamma_r} = \left(\frac{1000}{13.7} \right)^{0.33} \approx 4,$$

That is, adding roughly 4 units of baseline TbT effort is already enough to deter overtaking for this hardest-to-stabilize adjacent pair.

6. Limitations and Future Work

Our analysis provides a first step toward understanding the incentives induced by LLM benchmark leaderboards, but it relies on several simplifying assumptions. We discuss these limitations and future directions below.

Assumptions underlying the theoretical results. First, the tune-before-test guarantee relies on regularity assumptions on the post-effort score function (Assumption 4.2), motivated by controlled SFT settings. These assumptions may fail in more complex post-training pipelines, especially those involving reinforcement learning (Gao et al., 2023; Fernando et al., 2024). Extending the analysis to noisier or non-monotone dynamics is an important direction. Second, we model latent capability as commonly known to developers. In practice, developers may only have noisy information about their own and others’ models. A natural extension is a Bayesian contest model (Moldovanu & Sela, 2001), in which developers observe private signals, form beliefs about competitors, and choose post-training effort to maximize expected utility. Finally, we focus on pure-strategy equilibria in a static follower subgame. Real benchmarks are updated over time, and developers respond dynamically (Ott et al., 2022), which may induce different equilibrium behavior.

Multi-benchmark and multi-task settings. Our main analysis focuses on single-benchmark or independent multi-benchmark settings. In practice, developers often face multiple benchmarks or benchmark suites simultaneously, and must allocate shared post-training resources across them. Effort may concentrate on tasks or benchmarks that are easier to improve, more visible, or more strongly tied to downstream reward (Hardt, 2025). Tune-before-test can raise local overtaking costs within each benchmark, but may not stabilize global incentives when effort transfers across tasks, when benchmark scores are aggregated, or when benchmark suites evolve. Understanding such dynamic allocation problems with cross-task and cross-benchmark spillovers is an important future direction.

Practical limitations of tune-before-test. Tune-before-test is most applicable to centralized or platform-mediated leaderboards, where the designer can apply a standardized post-training procedure before evaluation. In decentralized settings, where developers self-report scores or run evaluations locally, implementing TbT would require additional standardization, logging, auditing, or selective reruns. TbT also imposes designer-side costs: applying it to every submitted model requires compute and infrastructure, with total cost growing at least linearly in the number of submissions. Practical deployments must therefore trade off incentive benefits against operational costs, especially for large-scale leaderboards or frontier models.

Acknowledgement

We thank Jiduan Wu, Vivian Nastl, Ana-Andreea Stoica, André Cruz, Ricardo Dominguez-Olmedo, Florian Dörner, Dingfan Chen, Tom Sühr, Nikhil Chandak, Mila Gorecki, Mina Remeli, Omri Ben-Dov, and anonymous reviewers from the ICLR AIMS workshop and ICML for the helpful discussions and feedback on this work.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

References

- Albalak, A., Elazar, Y., Xie, S. M., Longpre, S., Lambert, N., Wang, X., Muennighoff, N., Hou, B., Pan, L., Jeong, H., Raffel, C., Chang, S., Hashimoto, T., and Wang, W. Y. A survey on data selection for language models. *ArXiv*, abs/2402.16827, 2024. URL <https://api.semanticscholar.org/CorpusID:268032975>.
- Alon, T., Dobson, M., Procaccia, A., Talgam-Cohen, I., and Tucker-Foltz, J. Multiagent evaluation mechanisms. *Proc. AAAI Conference on Artificial Intelligence*, 34(02):1774–1781, Apr. 2020. URL <https://ojs.aaai.org/index.php/AAAI/article/view/5543>.
- Bisk, Y., Zellers, R., Bras, R. L., Gao, J., and Choi, Y. Piqa: Reasoning about physical commonsense in natural language. In *AAAI Conference on Artificial Intelligence*, 2019. URL <https://api.semanticscholar.org/CorpusID:208290939>.
- Blum, A. and Hardt, M. The ladder: A reliable leaderboard for machine learning competitions. In *International Conference on Machine Learning*, 2015. URL <https://api.semanticscholar.org/CorpusID:1493191>.
- Bordt, S., Srinivas, S., Boreiko, V., and von Luxburg, U. How much can we forget about data contamination? *ArXiv*, abs/2410.03249, 2024. URL <https://api.semanticscholar.org/CorpusID:273163321>.
- Brückner, M. and Scheffer, T. Stackelberg games for adversarial prediction problems. In *Proceedings of the 17th ACM SIGKDD international conference on Knowledge discovery and data mining*, pp. 547–555, 2011.
- Chen, Y., Wang, J., and Liu, Y. Learning to incentivize improvements from strategic agents. *Transactions on Machine Learning Research*, 2023. URL <https://openreview.net/forum?id=W98AEKQ38Y>.
- Clark, P., Cowhey, I., Etzioni, O., Khot, T., Sabharwal, A., Schoenick, C., and Tafjord, O. Think you have solved question answering? try arc, the ai2 reasoning challenge. *ArXiv*, abs/1803.05457, 2018. URL <https://api.semanticscholar.org/CorpusID:3922816>.
- Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., and Schulman, J. Training verifiers to solve math word problems. *ArXiv*, abs/2110.14168, 2021. URL <https://api.semanticscholar.org/CorpusID:239998651>.
- Connelly, B. L., Tihanyi, L., Crook, T. R., and Gangloff, K. A. Tournament theory: Thirty years of contests and competitions. *Journal of Management*, 40(1):16–47, 2014. doi: 10.1177/0149206313498902.
- Corchón, L. C. The theory of contests: a survey. *Review of Economic Design*, 11(2):69–100, 2007. URL <https://doi.org/10.1007/s10058-007-0032-5>.
- DatologyAI. Technical deep-dive: Curating our way to a state-of-the-art text dataset. <https://www.datologyai.com/blog/technical-deep-dive-curating-our-way-to-a-state-of-the-art-text-dataset>, November 2024.
- Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., and Fei-Fei, L. Imagenet: A large-scale hierarchical image database. *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 248–255, 2009. URL <https://api.semanticscholar.org/CorpusID:57246310>.
- Dominguez-Olmedo, R., Dörner, F. E., and Hardt, M. Training on the test task confounds evaluation and emergence. In *International Conference on Learning Representations*, volume 2025, pp. 68564–68594, 2025.
- Donoho, D. Data science at the singularity. *Harvard Data Science Review*, 6(1), 2024.
- Duda, R. O. and Hart, P. E. Pattern classification and scene analysis. In *A Wiley-Interscience publication*, 1974. URL <https://api.semanticscholar.org/CorpusID:12946615>.
- Dwork, C., Feldman, V., Hardt, M., Pitassi, T., Reingold, O., and Roth, A. Generalization in adaptive data analysis and holdout reuse. *ArXiv*, abs/1506.02629, 2015a. URL <https://api.semanticscholar.org/CorpusID:14762349>.

- Dwork, C., Feldman, V., Hardt, M., Pitassi, T., Reingold, O., and Roth, A. The reusable holdout: Preserving validity in adaptive data analysis. *Science*, 349:636 – 638, 2015b. URL <https://api.semanticscholar.org/CorpusID:15569600>.
- Einav, O. and Rosenfeld, N. A market for accuracy: Classification under competition. *arXiv preprint arXiv:2502.18052*, 2025.
- Fernando, H. D., Shen, H., Ram, P., Zhou, Y., Samulowitz, H., Baracaldo, N., and Chen, T. Mitigating forgetting in llm supervised fine-tuning and preference learning. 2024.
- Finnveden, L. Extrapolating gpt-n performance, 2020. URL <https://www.lesswrong.com/posts/k2SNji3jXaLGhBeYP/extrapolating-gpt-n-performance>.
- Fourrier, C., Habib, N., Lozovskaya, A., Szafer, K., and Wolf, T. Open llm leaderboard v2. https://huggingface.co/spaces/open-llm-leaderboard/open_llm_leaderboard, 2024.
- Fu, Q. and Wu, Z. Contests: Theory and topics. In *Oxford Research Encyclopedia of Economics and Finance*. Oxford University Press, 2019. doi: 10.1093/acrefore/9780190625979.013.440.
- Gao, L., Schulman, J., and Hilton, J. Scaling laws for reward model overoptimization. In *International Conference on Machine Learning*, pp. 10835–10866. PMLR, 2023.
- Garofolo, J. S., Lamel, L. F., Fisher, W. M., Fiscus, J. G., Pallett, D. S., and Dahlgren, N. L. DARPA TIMIT: Acoustic-phonetic continuous speech corpus cd-rom, NIST speech disc 1-1.1. Technical Report NISTIR 4930, National Institute of Standards and Technology, 1993.
- Glazer, E., Erdil, E., Besiroglu, T., Chicharro, D., Chen, E., Gunning, A., Olsson, C. F., Denain, J.-S., Ho, A., de Oliveira Santos, E., Järvinen, O., Barnett, M., Sandler, R., Vrzala, M., Sevilla, J., Ren, Q., Pratt, E., Levine, L., Barkley, G., Stewart, N., Grechuk, B., Grechuk, T., Enugandla, S. V., and Wildon, M. Frontiermath: A benchmark for evaluating advanced mathematical reasoning in ai. *ArXiv*, abs/2411.04872, 2024. URL <https://api.semanticscholar.org/CorpusID:273877467>.
- Goodfellow, I., Bengio, Y., and Courville, A. *Deep Learning*. MIT Press, 2016. <http://www.deeplearningbook.org>.
- Guha, E. K., Marten, R., Keh, S. S., Raoof, N., Smyrnis, G., Bansal, H., Nezhurina, M., Mercat, J.-P., Vu, T., Sprague, Z., Suvana, A., Feuer, B., Chen, L., Khan, Z., Frankel, E., Grover, S., Choi, C., Muennighoff, N., Su, S., Zhao, W., Yang, J., Pimpalgaonkar, S., Sharma, K., Ji, C. C.-J., Deng, Y., Pratt, S., Ramanujan, V., Saad-Falcon, J., Li, J., Dave, A., Albalak, A., Arora, K., Wulfe, B., Hegde, C., Durrett, G., Oh, S., Bansal, M., Gabriel, S., Grover, A., Chang, K.-W., Shankar, V., Gokaslan, A., Merrill, M. A., Hashimoto, T., Choi, Y., Jitsev, J., Heckel, R., Sathiamoorthy, M., Dimakis, A. G., and Schmidt, L. Openthoughts: Data recipes for reasoning models. *ArXiv*, abs/2506.04178, 2025. URL <https://api.semanticscholar.org/CorpusID:279154475>.
- Hardt, M. The emerging science of machine learning benchmarks. Online at <https://mlbenchmarks.org>, 2025. Manuscript.
- Hardt, M. and Recht, B. *Patterns, predictions, and actions: Foundations of machine learning*. Princeton University Press, 2022.
- Hardt, M., Megiddo, N., Papadimitriou, C., and Wootters, M. Strategic classification. In *Proceedings of the 2016 ACM conference on innovations in theoretical computer science*, pp. 111–122, 2016.
- He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2015. URL <https://api.semanticscholar.org/CorpusID:206594692>.
- Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2020.
- Hu, J. E., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., and Chen, W. Lora: Low-rank adaptation of large language models. *ArXiv*, abs/2106.09685, 2021. URL <https://api.semanticscholar.org/CorpusID:235458009>.
- Huan, M., Li, Y., Zheng, T., Xu, X., Kim, S., Du, M., Poovendran, R., Neubig, G., and Yue, X. Does math reasoning improve general llm capabilities? understanding transferability of llm reasoning. *ArXiv*, abs/2507.00432, 2025. URL <https://api.semanticscholar.org/CorpusID:280146966>.
- Jain, N., Han, K., Gu, A., Li, W.-D., Yan, F., Zhang, T., Wang, S., Solar-Lezama, A., Sen, K., and Stoica, I. Live-codebench: Holistic and contamination free evaluation of large language models for code. *ArXiv*, abs/2403.07974, 2024. URL <https://api.semanticscholar.org/CorpusID:268379413>.

- Jiang, M., Liu, K. Z., Zhong, M., Schaeffer, R., Ouyang, S., Han, J., and Koyejo, S. Investigating data contamination for pre-training language models. *ArXiv*, abs/2401.06059, 2024. URL <https://api.semanticscholar.org/CorpusID:266933004>.
- Kendall, M. G. A new measure of rank correlation. *Biometrika*, 30(1/2):81–93, 1938. ISSN 00063444. URL <http://www.jstor.org/stable/2332226>.
- Kleinberg, J. and Raghavan, M. How do classifiers induce agents to invest effort strategically? *ACM Transactions on Economics and Computation (TEAC)*, 8(4):1–23, 2020.
- Kornblith, S., Shlens, J., and Le, Q. V. Do better imagenet models transfer better? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 2661–2671, 2019.
- Krizhevsky, A., Sutskever, I., and Hinton, G. E. Imagenet classification with deep convolutional neural networks. *Communications of the ACM*, 60:84 – 90, 2012. URL <https://api.semanticscholar.org/CorpusID:195908774>.
- Kwiatkowski, T., Palomaki, J., Redfield, O., Collins, M., Parikh, A. P., Alberti, C., Epstein, D., Polosukhin, I., Devlin, J., Lee, K., Toutanova, K., Jones, L., Kellecey, M., Chang, M.-W., Dai, A. M., Uszkoreit, J., Le, Q. V., and Petrov, S. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466, 2019. URL <https://api.semanticscholar.org/CorpusID:86611921>.
- LeCun, Y. and Cortes, C. MNIST handwritten digit database. <http://yann.lecun.com/exdb/mnist/>, 2010. URL <http://yann.lecun.com/exdb/mnist/>.
- Li, J., Fang, A., Smyrnis, G., Ivgi, M., Jordan, M., Gadre, S. Y., Bansal, H., Guha, E. K., Keh, S. S., Arora, K., Garg, S., Xin, R., Muennighoff, N., Heckel, R., Mercat, J.-P., Chen, M., Gururangan, S., Wortsman, M., Albalak, A., Bitton, Y., Nezhurina, M., Abbas, A., Hsieh, C.-Y., Ghosh, D., Gardner, J., Kilian, M., Zhang, H., Shao, R., Pratt, S., Sanyal, S., Ilharco, G., Daras, G., Marathe, K., Gokaslan, A., Zhang, J., athi Chandu, K., Nguyen, T., Vasiljevic, I., Kakade, S. M., Song, S., Sanghavi, S., Faghri, F., Oh, S., Zettlemoyer, L. S., Lo, K., El-Nouby, A., Pouransari, H., Toshev, A., Wang, S., Groeneveld, D., Soldani, L., Koh, P. W., Jitsev, J., Kollar, T., Dimakis, A. G., Carmon, Y., Dave, A., Schmidt, L., and Shankar, V. Datacomp-lm: In search of the next generation of training sets for language models. *ArXiv*, abs/2406.11794, 2024. URL <https://api.semanticscholar.org/CorpusID:270560330>.
- Liang, P., Bommasani, R., Lee, T., Tsipras, D., Soylu, D., Yasunaga, M., Zhang, Y., Narayanan, D., Wu, Y., Kumar, A., Newman, B., Yuan, B., Yan, B., Zhang, C., Cosgrove, C., Manning, C. D., R’e, C., Acosta-Navas, D., Hudson, D. A., Zelikman, E., Durmus, E., Ladhak, F., Rong, F., Ren, H., Yao, H., Wang, J., Santhanam, K., Orr, L. J., Zheng, L., Yuksekgonul, M., Suzgun, M., Kim, N. S., Guha, N., Chatterji, N. S., Khattab, O., Henderson, P., Huang, Q., Chi, R., Xie, S. M., Santurkar, S., Ganguli, S., Hashimoto, T., Icard, T. F., Zhang, T., Chaudhary, V., Wang, W., Li, X., Mai, Y., Zhang, Y., and Koreeda, Y. Holistic evaluation of language models. *Annals of the New York Academy of Sciences*, 1525:140 – 146, 2023. URL <https://api.semanticscholar.org/CorpusID:253553585>.
- Liao, T. Are we learning yet? a meta review of evaluation failures across machine learning. In *NeurIPS Datasets and Benchmarks*, 2021. URL <https://api.semanticscholar.org/CorpusID:244907059>.
- Liberman, M. Reproducible research and the common task method, 2015. URL <https://www.simonsfoundation.org/lecture/reproducible-research-and-the-common-task-method>.
- Liu, L. T., Garg, N., and Borgs, C. Strategic ranking. In *International Conference on Artificial Intelligence and Statistics*, pp. 2489–2518. PMLR, 2022.
- Loshchilov, I. and Hutter, F. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2017. URL <https://api.semanticscholar.org/CorpusID:53592270>.
- Mangrulkar, S., Gugger, S., Debut, L., Belkada, Y., Paul, S., Bossan, B., and Tietz, M. PEFT: State-of-the-art parameter-efficient fine-tuning methods. <https://github.com/huggingface/peft>, 2022.
- Manheim, D. and Garrabrant, S. Categorizing variants of goodhart’s law. *ArXiv*, abs/1803.04585, 2018. URL <https://api.semanticscholar.org/CorpusID:4715794>.
- Mania, H., Miller, J., Schmidt, L., Hardt, M., and Recht, B. Model similarity mitigates test set overuse. *ArXiv*, abs/1905.12580, 2019. URL <https://api.semanticscholar.org/CorpusID:168169971>.
- Marchesi, A. *Leadership Games: Multiple Followers, Multiple Leaders, and Perfection*, pp. 107–118. Springer International Publishing, Cham, 2021. ISBN 978-3-030-62476-7. doi: 10.1007/978-3-030-62476-7_10. URL https://doi.org/10.1007/978-3-030-62476-7_10.

- Miller, J., Krauth, K., Recht, B., and Schmidt, L. The effect of natural distribution shift on question answering models. *ArXiv*, abs/2004.14444, 2020a. URL <https://api.semanticscholar.org/CorpusID:216867120>.
- Miller, J., Milli, S., and Hardt, M. Strategic classification is causal modeling in disguise. In *International Conference on Machine Learning*, pp. 6917–6926. PMLR, 2020b.
- Moldovanu, B. and Sela, A. The optimal allocation of prizes in contests. *American Economic Review*, 91(3):542–558, 2001.
- Nash, J. F. Equilibrium points in n-person games. *Proceedings of the National Academy of Sciences*, 36(1):48–49, 1950.
- Ott, S., Barbosa-Silva, A., Blagec, K., Brauner, J., and Samwald, M. Mapping global dynamics of benchmark creation and saturation in artificial intelligence. *Nature Communications*, 2022.
- Owen, D. How predictable is language model benchmark performance? *arXiv preprint*, arXiv:2401.04757, 2024. URL <https://arxiv.org/abs/2401.04757>.
- Pal, A., Umapathi, L. K., and Sankarasubbu, M. Medmcqa : A large-scale multi-subject multi-choice dataset for medical domain question answering. In *ACM Conference on Health, Inference, and Learning*, 2022. URL <https://api.semanticscholar.org/CorpusID:247763070>.
- Patwardhan, T., Dias, R., Proehl, E., Kim, G., Wang, M., Watkins, O., Fishman, S. P., Aljube, M., Thacker, P., Fauconnet, L., Kim, N. S., Chao, P., Miserendino, S., Chabot, G., Li, D., Sharman, M., Barr, A., Glaese, A., and Tworek, J. Gdpval: Evaluating ai model performance on real-world economically valuable tasks. *ArXiv*, abs/2510.04374, 2025. URL <https://api.semanticscholar.org/CorpusID:281843768>.
- Phan, L., Gatti, A., Han, Z., Li, N., Hu, J., Zhang, H., Zhang, C. B. C., Shaaban, M., Ling, J., Shi, S., et al. Humanity’s last exam. *arXiv preprint arXiv:2501.14249*, 2025.
- Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., Zhou, Y., Li, W., and Liu, P. J. Exploring the limits of transfer learning with a unified text-to-text transformer. *J. Mach. Learn. Res.*, 21:140:1–140:67, 2019. URL <https://api.semanticscholar.org/CorpusID:204838007>.
- Recht, B., Roelofs, R., Schmidt, L., and Shankar, V. Do imagenet classifiers generalize to imagenet? In *International Conference on Machine Learning*, 2019. URL <https://api.semanticscholar.org/CorpusID:67855879>.
- Rosenfeld, N. Strategic ml: How to learn with data that “behaves”. In *Proceedings of the 17th ACM International Conference on Web Search and Data Mining (WSDM)*, pp. 1128–1131. ACM, 2024.
- Ruan, Y., Maddison, C. J., and Hashimoto, T. B. Observational scaling laws and the predictability of language model performance. *Advances in Neural Information Processing Systems*, 37:15841–15892, 2024.
- Russakovsky, O., Deng, J., Su, H., Krause, J., Satheesh, S., Ma, S., Huang, Z., Karpathy, A., Khosla, A., Bernstein, M. S., Berg, A. C., and Fei-Fei, L. Imagenet large scale visual recognition challenge. *International Journal of Computer Vision*, 115:211 – 252, 2014. URL <https://api.semanticscholar.org/CorpusID:2930547>.
- Sakaguchi, K., Bras, R. L., Bhagavatula, C., and Choi, Y. Winogrande: An adversarial winograd schema challenge at scale, 2019. URL <https://arxiv.org/abs/1907.10641>.
- Salaudeen, O. and Hardt, M. Imagenot: A contrast with imagenet preserves model rankings. *ArXiv*, abs/2404.02112, 2024. URL <https://api.semanticscholar.org/CorpusID:268857319>.
- Salaudeen, O., Reuel, A., Ahmed, A. M., Bedi, S., Robertson, Z., Sundar, S., Domingue, B., Wang, A., and Koyejo, O. Measurement to meaning: A validity-centered framework for ai evaluation. *ArXiv*, abs/2505.10573, 2025. URL <https://api.semanticscholar.org/CorpusID:278715024>.
- Sang, E. T. K. and Meulder, F. D. Introduction to the conll-2003 shared task: Language-independent named entity recognition. In *Conference on Computational Natural Language Learning*, 2003. URL <https://api.semanticscholar.org/CorpusID:2470716>.
- Sap, M., Rashkin, H., Chen, D., LeBras, R., and Choi, Y. Socialliqa: Commonsense reasoning about social interactions, 2019. URL <https://arxiv.org/abs/1904.09728>.
- Schaeffer, R., Miranda, B., and Koyejo, O. Are emergent abilities of large language models a mirage? *ArXiv*, abs/2304.15004, 2023. URL <https://api.semanticscholar.org/CorpusID:258418299>.
- Singh, S., Nan, Y., Wang, A., D’Souza, D., Kapoor, S., Üstün, A., Koyejo, S., Deng, Y., Longpre, S., Smith, N. A., et al. The leaderboard illusion. *arXiv preprint arXiv:2504.20879*, 2025.

- Sommer, Y., Hikri, I., Amit, L., and Rosenfeld, N. Learning classifiers that induce markets. In *Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)*, 2025.
- Talmor, A., Herzig, J., Lourie, N., and Berant, J. Commonsenseqa: A question answering challenge targeting commonsense knowledge. *ArXiv*, abs/1811.00937, 2019. URL <https://api.semanticscholar.org/CorpusID:53296520>.
- Topkis, D. M. *Supermodularity and Complementarity*. Princeton University Press, 1998. ISBN 9780691032443. URL <http://www.jstor.org/stable/j.ctt7s83q>.
- Touvron, H., Martin, L., Stone, K. R., Albert, P., Almahairi, A., Babaei, Y., Bay, Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., Bikel, D. M., Blecher, L., Ferrer, C. C., Chen, M., Cucurull, G., Esiobu, D., Fernandes, J., Fu, J., Fu, W., Fuller, B., Gao, C., Goswami, V., Goyal, N., Hartshorn, A. S., Hosseini, S., Hou, R., Inan, H., Kardaş, M., Kerkez, V., Khabsa, M., Kloumann, I. M., Korenev, A., Koura, P. S., Lachaux, M.-A., Lavril, T., Lee, J., Liskovich, D., Lu, Y., Mao, Y., Martinet, X., Mihaylov, T., Mishra, P., Molybog, I., Nie, Y., Poulton, A., Reizenstein, J., Rungta, R., Saladi, K., Schelten, A., Silva, R., Smith, E. M., Subramanian, R., Tan, X., Tang, B., Taylor, R., Williams, A., Kuan, J. X., Xu, P., Yan, Z., Zarov, I., Zhang, Y., Fan, A., Kambadur, M. H. M., Narang, S., Rodriguez, A., Stojnic, R., Edunov, S., and Scialom, T. Llama 2: Open foundation and fine-tuned chat models. *ArXiv*, abs/2307.09288, 2023. URL <https://api.semanticscholar.org/CorpusID:259950998>.
- Von Stengel, B. and Zamir, S. Leadership games with convex strategy sets. *Games and Economic Behavior*, 69 (2):446–457, 2010.
- Wei, J., Bosma, M., Zhao, V. Y., Guu, K., Yu, A. W., Lester, B., Du, N., Dai, A. M., and Le, Q. V. Finetuned language models are zero-shot learners. *arXiv preprint arXiv:2109.01652*, 2021.
- Yadav, C. and Bottou, L. Cold case: The lost mnist digits. In *Neural Information Processing Systems*, 2019. URL <https://api.semanticscholar.org/CorpusID:166227957>.
- Yang, Q. A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Li, C., Liu, D., Huang, F., Dong, G., Wei, H., Lin, H., Yang, J., Tu, J., Zhang, J., Yang, J., Yang, J., Zhou, J., Lin, J., Dang, K., Lu, K., Bao, K., Yang, K., Yu, L., Li, M., Xue, M., Zhang, P., Zhu, Q., Men, R., Lin, R., Li, T., Xia, T., Ren, X., Ren, X., Fan, Y., Su, Y., Zhang, Y.-C., Wan, Y., Liu, Y., Cui, Z., Zhang, Z., Qiu, Z., Quan, S., and Wang, Z. Qwen2.5 technical report. *ArXiv*, abs/2412.15115, 2024. URL <https://api.semanticscholar.org/CorpusID:274859421>.
- Yang, S., Chiang, W.-L., Zheng, L., Gonzalez, J., and Stoica, I. Rethinking benchmark and contamination for language models with rephrased samples. *ArXiv*, abs/2311.04850, 2023. URL <https://api.semanticscholar.org/CorpusID:265050721>.
- Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., and Choi, Y. Hellaswag: Can a machine really finish your sentence? In *Annual Meeting of the Association for Computational Linguistics*, 2019a. URL <https://api.semanticscholar.org/CorpusID:159041722>.
- Zellers, R., Holtzman, A., Bisk, Y., Farhadi, A., and Choi, Y. Hellaswag: Can a machine really finish your sentence? *arXiv preprint arXiv:1905.07830*, 2019b.
- Zhang, G. and Hardt, M. Inherent trade-offs between diversity and stability in multi-task benchmarks. *ArXiv*, abs/2405.01719, 2024. URL <https://api.semanticscholar.org/CorpusID:269587753>.
- Zhang, G., Dominguez-Olmedo, R., and Hardt, M. Train-before-test harmonizes language model rankings. *arXiv preprint arXiv:2507.05195*, 2025.
- Zhou, J., Lu, T., Mishra, S., Brahma, S., Basu, S., Luan, Y., Zhou, D., and Hou, L. Instruction-following evaluation for large language models. *ArXiv*, abs/2311.07911, 2023a. URL <https://api.semanticscholar.org/CorpusID:265157752>.
- Zhou, K., Zhu, Y., Chen, Z., Chen, W., Zhao, W. X., Chen, X., Lin, Y., Wen, J.-R., and Han, J. Don’t make your llm an evaluation benchmark cheater. *ArXiv*, abs/2311.01964, 2023b. URL <https://api.semanticscholar.org/CorpusID:265019021>.

A. Primary Notation Table

Symbol	Description
$\theta \in \mathbb{R}_{\geq 0}$	one-dimensional latent model capability
$e \in \mathbb{R}_{\geq 0}$	benchmark-specific training effort
$v(\theta, e) \in [0, 1]$	post-effort score (benchmark performance)
$c(e)$	cost of exerting effort e
$(R_j)_{j=1}^n$	rank-based reward scheme: the model ranked j receives reward R_j
$\Delta^{ibt} \in \mathbb{R}_{\geq 0}$	tune-before-test (TbT) baseline chosen by the leaderboard designer
$s_r(\Delta^{ibt}) := v(\theta_r, \Delta^{ibt})$	baseline score of model r under TbT level Δ^{ibt}
$e^{\text{req}}(s; \theta)$	minimal effort required for a model with capability θ to reach score s , i.e., $e^{\text{req}}(s; \theta) := \inf\{e \geq 0 : v(\theta, e) \geq s\}$
$e_r^{\text{req}}(\Delta^{ibt})$	minimal additional effort required for model r to overtake model $r-1$ at TbT level Δ^{ibt} , i.e., $e_r^{\text{req}}(\Delta^{ibt}) := \inf\{e \geq 0 : v(\theta_r, \Delta^{ibt} + e) > s_{r-1}(\Delta^{ibt})\}$
$\sigma : \mathbb{R} \rightarrow [0, 1]$	logit link function
$L(\theta)$	lower attainable performance level for a model with capability θ
$U(\theta)$	upper attainable performance level for a model with capability θ
$\tilde{v}(\theta, e) := \frac{v(\theta, e) - L(\theta)}{U(\theta) - L(\theta)} \in [0, 1]$	normalized post-effort score
$\alpha(\theta)$	logit of the baseline performance when $e = 0$
$\beta(\theta) > 0$	coefficient governing how effectively extra effort translates into performance
$\kappa > 0$	constant lower bound on the marginal cost of effort, i.e., $c(e) \geq \kappa e$
$\rho_r := (R_{r-1} - R_r)/\kappa$	effective reward gap (in effort units) for model r
$\lambda_r := \rho_r / e_r^{\text{req}}(0)$	effective incentive parameter
$\gamma_r := \beta(\theta_r) / \beta(\theta_{r-1}) \leq 1$	learning-rate ratio for the pair $(r-1, r)$
Δ_r^{ibt*}	stabilizing TbT threshold for the adjacent pair $(r-1, r)$
$\Delta^{ibt*} := \max_{r \in \{2, \dots, n\}} \Delta_r^{ibt*}$	global stabilizing TbT threshold

Table 1. Primary notation.

B. Missing Proof in Section 3

Equivalence of heterogeneous separable costs to a homogeneous same-family cost

Proposition B.1 (Equivalence of heterogeneous separable costs to a homogeneous same-family cost). *Fix a common cost shape $c : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ that is continuous, strictly increasing, and satisfies $c(0) = 0$. Consider each model i chooses effort $e_i \in \mathbb{R}_+$, is ranked by a score $v(\theta_i, e_i)$, and has a heterogeneous but multiplicatively separable cost*

$$C_i(e_i; \theta_i) = \gamma_i c(e_i) \quad \text{with } \gamma_i > 0.$$

Then there exists an equivalent game in which every model developer chooses $z_i \in \mathbb{R}_+$, all model developers share the same cost function of the same form $c(\cdot)$ (i.e., homogeneous cost $c(z_i)$), and the only change is a type-dependent relabeling of the score:

$$\hat{v}(\theta_i, z_i) := v\left(\theta_i, c^{-1}\left(\frac{c(z_i)}{\gamma_i}\right)\right).$$

Specifically, the map

$$\Phi_i : e_i \leftrightarrow z_i := c^{-1}(\gamma_i c(e_i))$$

is a bijection for each i , and for every profile $(e_j)_j$ with image $(z_j)_j$ we have

$$R_{\text{Rank}(v(\theta_i, e_i))} - \gamma_i c(e_i) = R_{\text{Rank}(\hat{v}(\theta_i, z_i))} - c(z_i) \quad \text{for all } i.$$

Consequently, the two games induce identical rankings, payoffs, best responses, and Nash equilibria up to the one-to-one reparametrization.

Proof. Because c is strictly increasing, c^{-1} exists and is strictly increasing. For each i , define the type-wise bijection $\Phi_i(e) := c^{-1}(\gamma_i c(e))$, with inverse

$$\Phi_i^{-1}(z) = c^{-1}\left(\frac{c(z)}{\gamma_i}\right).$$

Under Φ_i , the heterogeneous cost transforms as

$$\gamma_i c(e_i) = c(\Phi_i(e_i)) = c(z_i),$$

so all model developers share the same cost function $c(\cdot)$ in the z -parametrization. Define $\hat{v}(\theta_i, z_i) := v(\theta_i, \Phi_i^{-1}(z_i)) = v(\theta_i, c^{-1}(c(z_i)/\gamma_i))$. Then for any profile $(e_j)_j$ with the reparametrization variable $(z_j)_j$, we have:

$$R_{\text{Rank}(v(\theta_i, e_i))} - \gamma_i c(e_i) = R_{\text{Rank}(v(\theta_i, \Phi_i^{-1}(z_i)))} - c(z_i) = R_{\text{Rank}(\hat{v}(\theta_i, z_i))} - c(z_i),$$

so the utilities are preserved. Hence, the best responses and equilibria correspond one-to-one via Φ , establishing the equivalence between the two settings. $\square$

C. Proofs for Section 4

Proof for Proposition 4.3

Proof. Fix any $\Delta^{ibt} \geq 0$ and let $\mathbf{e}^*$ be a PNE of the induced follower game. Suppose for contradiction that there exist models i, j such that $\theta_i > \theta_j$ but

$$v_i^* := v(\theta_i, \Delta^{ibt} + e_i^*) < v(\theta_j, \Delta^{ibt} + e_j^*) =: v_j^*. \quad (7)$$

Counterfactual efforts. For any target score $s \in [0, 1]$, recall the minimal additional effort (beyond Δ^{ibt}) needed for capability θ to reach s is:

$$e^{\text{req}}(s; \theta, \Delta^{ibt}) := \inf\{e \geq 0 : v(\theta, \Delta^{ibt} + e) \geq s\}.$$

Define the counterfactual efforts

$$\tilde{e}_i := e^{\text{req}}(v_j^*; \theta_i, \Delta^{ibt}), \quad \tilde{e}_j := e^{\text{req}}(v_i^*; \theta_j, \Delta^{ibt}).$$

Then by definition,

$$v(\theta_i, \Delta^{ibt} + \tilde{e}_i) \geq v_j^*, \quad v(\theta_j, \Delta^{ibt} + \tilde{e}_j) \geq v_i^*.$$

Moreover, since $v_j^* > v_i^*$ and $v(\theta, \cdot)$ is nondecreasing in effort (C2), we have $\tilde{e}_i > \tilde{e}_j$ and $\tilde{e}_j \leq e_j^*$.

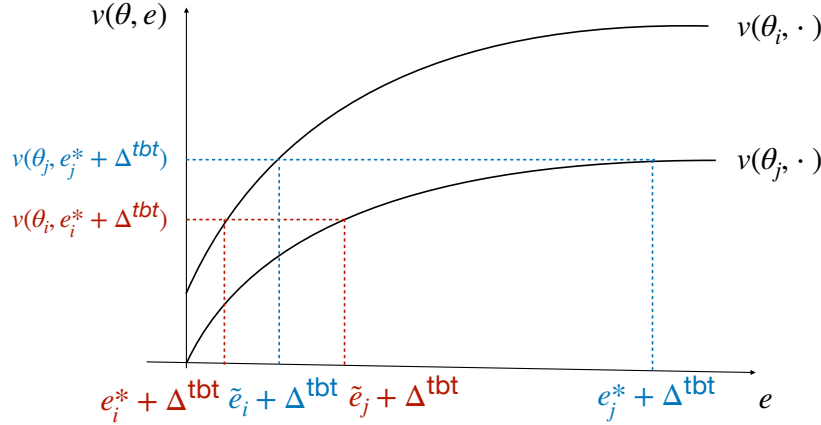


Figure 2. Illustration of the notations and functions used in the proof of Proposition 4.3. The post-effort score function $v(\theta, e)$ satisfies the conditions in Assumption 4.2. Here, e_i^* and e_j^* denote the equilibrium efforts of model developers i and j , respectively. The counterfactual efforts $\tilde{e}_i$ and $\tilde{e}_j$ are defined as the efforts each model developer would need to match the other model developer's post-effort score given a TbT effort level Δ^{tbt} , i.e., $v(\theta_i, \tilde{e}_i + \Delta^{tbt}) = v(\theta_j, e_j^* + \Delta^{tbt})$, $v(\theta_j, \tilde{e}_j + \Delta^{tbt}) = v(\theta_i, e_i^* + \Delta^{tbt})$.

Effort-gap comparison. Let

$$\Delta_{\theta_i, \theta_j}(s) := e^{\text{req}}(s; \theta_j, \Delta^{tbt}) - e^{\text{req}}(s; \theta_i, \Delta^{tbt})$$

denote the effort gap required for the lower-capability model j to achieve score s relative to the higher-capability model i (at TbT level Δ^{tbt}). By (C3), $\Delta_{\theta_i, \theta_j}(s)$ is (weakly) nondecreasing in s . Applying this with $s = v_j^* > v_i^*$ gives

$$\Delta_{\theta_i, \theta_j}(v_j^*) \geq \Delta_{\theta_i, \theta_j}(v_i^*),$$

i.e.,

$$e^{\text{req}}(v_j^*; \theta_j, \Delta^{tbt}) - e^{\text{req}}(v_j^*; \theta_i, \Delta^{tbt}) \geq e^{\text{req}}(v_i^*; \theta_j, \Delta^{tbt}) - e^{\text{req}}(v_i^*; \theta_i, \Delta^{tbt}).$$

Using $e^{\text{req}}(v_i^*; \theta_i, \Delta^{tbt}) = e_i^*$ and $e^{\text{req}}(v_j^*; \theta_j, \Delta^{tbt}) = e_j^*$, and the definitions $\tilde{e}_i := e^{\text{req}}(v_j^*; \theta_i, \Delta^{tbt})$ and $\tilde{e}_j := e^{\text{req}}(v_i^*; \theta_j, \Delta^{tbt})$, this becomes

$$\tilde{e}_i - e_i^* \leq e_j^* - \tilde{e}_j. \quad (8)$$

By convexity and monotonicity of c (Assumption 4.1), Equation (8) implies

$$c(\tilde{e}_i) - c(e_i^*) \leq c(e_j^*) - c(\tilde{e}_j). \quad (9)$$

Nash inequalities. Since $\mathbf{e}^*$ is a PNE, neither model can gain by deviating unilaterally. Let $\mathbf{v}^* = (v_1^*, \dots, v_n^*)$ be the equilibrium score profile and let $r_k(\cdot)$ denote the rank of model k under a given score profile (with an arbitrary but fixed deterministic tie-breaking rule). Then

$$R_{r_i(\mathbf{v}^*)} - c(e_i^*) \geq R_{r_i(v(\theta_i, \Delta^{tbt} + \tilde{e}_i), \mathbf{v}_{-i}^*)} - c(\tilde{e}_i),$$

$$R_{r_j(\mathbf{v}^*)} - c(e_j^*) \geq R_{r_j(v(\theta_j, \Delta^{tbt} + \tilde{e}_j), \mathbf{v}_{-j}^*)} - c(\tilde{e}_j).$$

Rearranging terms gives

$$R_{r_i(v(\theta_i, \Delta^{tbt} + \tilde{e}_i), \mathbf{v}_{-i}^*)} - R_{r_i(\mathbf{v}^*)} \leq c(\tilde{e}_i) - c(e_i^*), \quad (10)$$

$$R_{r_j(\mathbf{v}^*)} - R_{r_j(v(\theta_j, \Delta^{tbt} + \tilde{e}_j), \mathbf{v}_{-j}^*)} \geq c(e_j^*) - c(\tilde{e}_j). \quad (11)$$

Here Equation (10) is the reward increase when i raises its effort to (at least) match j 's score v_j^* , and Equation (11) is the reward decrease when j lowers its effort to (at least) match i 's score v_i^* .

Reward comparison via strict overtaking. Fix any $\varepsilon > 0$, and define the ε -overtake effort for model i :

$$\tilde{e}_i^\varepsilon := \inf\{e \geq 0 : v(\theta_i, \Delta^{ibt} + e) \geq v_j^* + \varepsilon\}.$$

Then $v(\theta_i, \Delta^{ibt} + \tilde{e}_i^\varepsilon) > v_j^*$, so model i strictly outranks model j under *any* deterministic tie-breaking rule. Hence the reward increase of model i from deviating to $\tilde{e}_i^\varepsilon$ is at least the reward decrease of model j when it deviates to achieve v_j^* :

$$R_{r_i}(v(\theta_i, \Delta^{ibt} + \tilde{e}_i^\varepsilon), \mathbf{v}_{-i}^*) - R_{r_i}(\mathbf{v}^*) \geq R_{r_j}(\mathbf{v}^*) - R_{r_j}(v(\theta_j, \Delta^{ibt} + \tilde{e}_j), \mathbf{v}_{-j}^*). \quad (12)$$

Combining (10), (11), and (12) yields

$$c(\tilde{e}_i^\varepsilon) - c(e_i^*) \geq c(e_j^*) - c(\tilde{e}_j).$$

Letting $\varepsilon \downarrow 0$ and using continuity of $v(\theta, \cdot)$ and $c(\cdot)$ gives

$$c(\tilde{e}_i) - c(e_i^*) \geq c(e_j^*) - c(\tilde{e}_j).$$

Together with (9), we must have

$$c(\tilde{e}_i) - c(e_i^*) = c(e_j^*) - c(\tilde{e}_j).$$

However, since $v_i^* < v_j^*$ by assumption, model i can strictly improve its rank by exerting effort slightly above $\tilde{e}_i$ (i.e., achieving $v_j^* + \varepsilon$), which yields a *strictly* higher reward whenever rewards are strictly decreasing in rank. This would force a strict inequality in the Nash bound for i , contradicting equality.

Therefore Equation (7) is impossible, and for any $\theta_i > \theta_j$ we must have

$$v(\theta_i, \Delta^{ibt} + e_i^*) \geq v(\theta_j, \Delta^{ibt} + e_j^*).$$

In particular, equilibrium scores are capability-consistent (up to ties). □

Proof for Proposition 4.5

Proof. Fix any tune-before-test adjustment level $\Delta^{ibt} \geq 0$. At the all-zero *additional-effort* profile $\mathbf{e} = (0, \dots, 0)$, the realized scores are $s_i(\Delta^{ibt}) = v(\theta_i, \Delta^{ibt})$ for $i = 1, \dots, n$. Since models are indexed so that $\theta_1 > \theta_2 > \dots > \theta_n$ and $v(\theta, e)$ is increasing in θ (C1), we have $s_1(\Delta^{ibt}) > s_2(\Delta^{ibt}) > \dots > s_n(\Delta^{ibt})$. Hence the payoff of model k at this profile is $U_k(0; \mathbf{0}_{-k}, \Delta^{ibt}) = R_k$ for $k = 1, \dots, n$.

Fix a model $r \in \{2, \dots, n\}$. Consider any unilateral deviation $e_r' > 0$ by model r . Because rewards depend only on ranks and c is nondecreasing, among all deviations that improve model r 's rank, the cheapest such deviation is to *just overtake* some model currently above it, i.e., to move from rank r to some rank $k < r$ by achieving a score strictly greater than $s_k(\Delta^{ibt})$. Define the minimal additional effort needed for model r to strictly beat score $s_k(\Delta^{ibt})$ as

$$e^{\text{req}}(s_k(\Delta^{ibt}); \theta_r, \Delta^{ibt}) := \inf\{e \geq 0 : v(\theta_r, \Delta^{ibt} + e) > s_k(\Delta^{ibt})\}.$$

Such a deviation yields utility

$$R_k - c(e^{\text{req}}(s_k(\Delta^{ibt}); \theta_r, \Delta^{ibt})).$$

Therefore, $e_r = 0$ is a best response for model r at the all-zero profile if and only if

$$R_r \geq R_k - c(e^{\text{req}}(s_k(\Delta^{ibt}); \theta_r, \Delta^{ibt})) \quad \text{for all } k < r,$$

equivalently,

$$c(e^{\text{req}}(s_k(\Delta^{ibt}); \theta_r, \Delta^{ibt})) \geq R_k - R_r \quad \text{for all } k < r.$$

and we shorthand $e_r^{\text{req}}(\Delta^{ibt}) := e^{\text{req}}(s_{r-1}(\Delta^{ibt}); \theta_r, \Delta^{ibt})$.

Finally, since overtaking a higher-ranked model requires weakly more effort than overtaking the adjacent model $r - 1$, it suffices to check the adjacent deviation $k = r - 1$, yielding Equation (4). □

Proof for Theorem 4.6

Proof. We first show that if a PNE exists, it must be the all-zero additional-effort profile; we then show that under the stated condition, the all-zero profile cannot be a PNE.

Fix $\Delta^{ibt} \geq 0$ and suppose $\mathbf{e}^*$ is a PNE. By Proposition 4.3, post-effort scores are weakly ordered by capability at equilibrium: if $\theta_i > \theta_j$, then $v(\theta_i, \Delta^{ibt} + e_i^*) \geq v(\theta_j, \Delta^{ibt} + e_j^*)$. Under deterministic tie-breaking that favors higher capability, this implies that model n (the lowest capability θ_n) is last-ranked at effort level $\mathbf{e}^*$. Since at any effort profile, the last-ranked model receives reward R_n regardless of its own effort, thus model n can weakly improve its utility by reducing its effort to zero: its reward cannot decrease, while its cost weakly decreases, since c is nondecreasing and $c(0) = 0$. Therefore, in any PNE, we must have $e_n^* = 0$.

Now consider model $n - 1$. Given $e_n^* = 0$, if model $n - 1$ sets effort to zero as well, its score becomes $v(\theta_{n-1}, \Delta^{ibt})$, which is weakly higher than $v(\theta_n, \Delta^{ibt})$ by the assumption that $v(\theta, e)$ is monotonic in capability (C1). Under the same tie-breaking rule, model $n - 1$ cannot fall below model n by choosing zero effort, and its cost decreases. Hence $e_{n-1}^* = 0$ as well. Proceeding inductively, we obtain $e_i^* = 0$ for all i , so any PNE must be $\mathbf{e}^* = (0, \dots, 0)$.

Next, we show that under the stated condition Equation (5), all-zero is not a PNE. At $\mathbf{e} = (0, \dots, 0)$, baseline scores are $s_i(\Delta^{ibt}) = v(\theta_i, \Delta^{ibt})$. If for some $r \in \{2, \dots, n\}$,

$$c(e_r^{\text{req}}(\Delta^{ibt})) < R_{r-1} - R_r,$$

then model r has a profitable deviation, namely that it can exert effort $\hat{e} = e_r^{\text{req}}(\Delta^{ibt})$, so that $v(\theta_r, \Delta^{ibt} + \hat{e}) > s_{r-1}(\Delta^{ibt})$, thereby overtaking model $r - 1$ and improving its reward from R_r to at least R_{r-1} . The resulting utility gain is at least $R_{r-1} - R_r$ while the incurred cost is $c(\hat{e})$, which is strictly smaller by assumption. Therefore, the deviation is profitable, so the all-zero profile is not a PNE. As a result, no PNE exists in this case. $\square$

D. Proof in Section 5

Proof for Proposition 5.1

Proof. Fix any pair (i, j) with $\theta_i > \theta_j$ and define the score gap at effort level $e \geq 0$ as $d_{ij}(e) := v(\theta_i, e) - v(\theta_j, e)$. By C1 ($\partial_\theta v > 0$), we have $d_{ij}(e) > 0$ for all e . Hence,

$$v^\infty(\theta_i) - v^\infty(\theta_j) = \lim_{e \rightarrow \infty} (v(\theta_i, e) - v(\theta_j, e)) = \lim_{e \rightarrow \infty} d_{ij}(e) > 0,$$

so $v^\infty(\theta)$ is increasing in θ . $\square$

Proof for Lemma 5.2

Proof. Fix $r \in \{2, \dots, n\}$ with $\theta_{r-1} > \theta_r$, recall the baseline score for model $r - 1$ with TbT adjustment level Δ^{ibt} is

$$s_{r-1}(\Delta^{ibt}) := v(\theta_{r-1}, \Delta^{ibt}).$$

By C2, $s_{r-1}(\Delta^{ibt})$ is non-decreasing in Δ^{ibt} .

Let $e^{\text{req}}(s; \theta) := \inf\{e \geq 0 : v(\theta, e) \geq s\}$ be the minimal *total* effort to reach score s . Define the (weak) catch-up effort for model r at baseline Δ^{ibt} by

$$\delta_r(\Delta^{ibt}) := e^{\text{req}}(s_{r-1}(\Delta^{ibt}); \theta_r) - \Delta^{ibt}.$$

Since $s_{r-1}(\Delta^{ibt})$ is achieved by capability θ_{r-1} at effort Δ^{ibt} , we also have $e^{\text{req}}(s_{r-1}(\Delta^{ibt}); \theta_{r-1}) = \Delta^{ibt}$, hence

$$\delta_r(\Delta^{ibt}) = e^{\text{req}}(s_{r-1}(\Delta^{ibt}); \theta_r) - e^{\text{req}}(s_{r-1}(\Delta^{ibt}); \theta_{r-1}).$$

By (C3), the effort gap $e^{\text{req}}(s; \theta_r) - e^{\text{req}}(s; \theta_{r-1})$ is nondecreasing in S ; composing with the nondecreasing map $\Delta^{ibt} \mapsto s_{r-1}(\Delta^{ibt})$ implies $\delta_r(\Delta^{ibt})$ is nondecreasing in Δ^{ibt} .

Finally, the strict “just-overtake” effort $e_r^{\text{req}}(\Delta^{ibt})$ is obtained by requiring $v(\theta_r, \Delta^{ibt} + e) > s_{r-1}(\Delta^{ibt})$ instead of $\geq$; by continuity of $v(\theta_r, \cdot)$ this differs only by an arbitrarily small $\varepsilon > 0$, so $e_r^{\text{req}}(\Delta^{ibt})$ is also nondecreasing in Δ^{ibt} .

Since c is nondecreasing, $c(e_r^{\text{req}}(\Delta^{ibt}))$ is nondecreasing as well. $\square$

Proof for Proposition 5.3

Proof. Given $0 \leq \Delta^{ibt}_1 < \Delta^{ibt}_2$, and assume the all-zero additional-effort profile $\mathbf{e} = (0, \dots, 0)$ is a PNE of the induced follower game under baseline Δ^{ibt}_1 , we know from Proposition 4.5 that $\forall r \in \{2, \dots, n\}$, we have

$$c(e_r^{\text{req}}(\Delta^{ibt}_1)) \geq R_{r-1} - R_r.$$

Then according to Lemma 5.2, the cost of "just-overtake" effort $c(e_r^{\text{req}}(\Delta^{ibt}))$ is monotonically increasing in Δ^{ibt} , which means:

$$c(e_r^{\text{req}}(\Delta^{ibt}_2)) \geq c(e_r^{\text{req}}(\Delta^{ibt}_1)) \geq R_{r-1} - R_r,$$

which implies that $\mathbf{e} = (0, \dots, 0)$ is also a PNE under Δ^{ibt}_2 . $\square$

Proof for Corollary 5.5

Proof. By definition of Δ^{ibt*} , for any $\Delta^{ibt} \geq \Delta^{ibt*}$ we have $c(e_r^{\text{req}}(\Delta^{ibt})) \geq R_{r-1} - R_r$ for $r = \{2, \dots, n\}$, so the all-zero profile is a PNE by Proposition 4.5. $\square$

Proof for Proposition 5.6

Proof. Since σ is strictly increasing, the catch-up condition $v(\theta_r, \Delta^{ibt} + \delta) \geq v(\theta_{r-1}, \Delta^{ibt})$ is equivalent to

$$\tilde{v}(\theta_r, \Delta^{ibt} + \delta) \geq \frac{v(\theta_{r-1}, \Delta^{ibt}) - L(\theta_r)}{U(\theta_r) - L(\theta_r)}.$$

Applying σ^{-1} and the scaling law for $\tilde{v}(\theta_r, \cdot)$, this becomes

$$\alpha(\theta_r) + \beta(\theta_r) \log(1 + \Delta^{ibt} + \delta) \geq \sigma^{-1} \left(\frac{v(\theta_{r-1}, \Delta^{ibt}) - L(\theta_r)}{U(\theta_r) - L(\theta_r)} \right).$$

Solving with equality yields the minimal additional effort:

$$1 + \Delta^{ibt} + \delta = \exp \left(\frac{1}{\beta(\theta_r)} \left[\sigma^{-1} \left(\frac{v(\theta_{r-1}, \Delta^{ibt}) - L(\theta_r)}{U(\theta_r) - L(\theta_r)} \right) - \alpha(\theta_r) \right] \right).$$

Define $e_r^{\text{req}}(\Delta^{ibt})$ to be this minimal δ , then subtracting $(1 + \Delta^{ibt})$ gives

$$e_r^{\text{req}}(\Delta^{ibt}) = \exp \left(\frac{1}{\beta(\theta_r)} \left[\sigma^{-1} \left(\frac{v(\theta_{r-1}, \Delta^{ibt}) - L(\theta_r)}{U(\theta_r) - L(\theta_r)} \right) - \alpha(\theta_r) \right] \right) - (1 + \Delta^{ibt}).$$

Using the scaling law for θ_{r-1} ,

$$\sigma^{-1}(\tilde{v}(\theta_{r-1}, \Delta^{ibt})) = \alpha(\theta_{r-1}) + \beta(\theta_{r-1}) \log(1 + \Delta^{ibt}),$$

and the definition of $\gamma_r = \beta(\theta_r)/\beta(\theta_{r-1})$, we can rewrite the leading term as

$$\exp \left(\frac{\bar{\alpha}_r(\Delta^{ibt})}{\beta(\theta_r)} \right) (1 + \Delta^{ibt})^{1/\gamma_r},$$

where $\bar{\alpha}_r(\Delta^{ibt}) = \sigma^{-1}(\tilde{v}(\theta_{r-1}, \Delta^{ibt})) - \alpha(\theta_r)$, yielding the stated expression.

With costs $c(e) \geq \kappa e$, any deviation that uses additional effort δ incurs cost at least $\kappa \delta$, so an adjacent-overtake deviation is unprofitable whenever $e_r^{\text{req}}(\Delta^{ibt}) \geq \rho_r$.

Finally, in the regime where the leading term in $e_r^{\text{req}}(\Delta^{ibt})$ dominates $(1 + \Delta^{ibt})$, the condition $e_r^{\text{req}}(\Delta^{ibt}) \geq \rho_r$ is well-approximated by

$$\exp \left(\frac{\bar{\alpha}_r(\Delta^{ibt})}{\beta(\theta_r)} \right) (1 + \Delta^{ibt})^{1/\gamma_r} \gtrsim \rho_r.$$

Treating $\bar{\alpha}_r(\Delta^{ibt})$ as approximately constant over the relevant range (or evaluating it at $\Delta^{ibt} = 0$ to obtain a conservative rule of thumb) gives

$$1 + \Delta^{ibt} \gtrsim \left(\frac{\rho_r}{\exp(\bar{\alpha}_r(0)/\beta(\theta_r))} \right)^{\gamma_r} = \left(\frac{\rho_r}{e_r^{\text{req}}(0) + 1} \right)^{\gamma_r},$$

which implies the stated scaling. Since $\lambda_r = \rho_r / e_r^{\text{req}}(0)$, we also have $\Delta^{ibt*}_r = O(\lambda_r^{\gamma_r})$ up to a constant factor depending only on $e_r^{\text{req}}(0)$. $\square$

Estimating $e_r^{\text{req}}(0)$, γ_r , and λ_r from the post-training trajectories. For each model i , we fit the generalized scaling form $\sigma^{-1}(\tilde{v}_i(e)) = \alpha_i + \beta_i \log(1 + e)$ by linear regression of $\sigma^{-1}(\tilde{v}_i)$ on $\log(1 + e)$ over the observed training range, yielding $(\hat{\alpha}_i, \hat{\beta}_i)$. For an adjacent pair $(r-1, r)$ we set

$$\hat{\gamma}_r := \frac{\hat{\beta}_r}{\hat{\beta}_{r-1}}.$$

The baseline catch-up difficulty $e_r^{\text{req}}(0)$ is defined as the minimal additional effort at $\Delta^{ibt} = 0$ required for model r to match model $r-1$; under the fitted scaling law,

$$\widehat{e_r^{\text{req}}(0)} = \exp\left(\frac{\hat{\alpha}_{r-1} - \hat{\alpha}_r}{\hat{\beta}_r}\right) - 1.$$

Given a reward gap (converted to effort units) $\rho_r = (R_{r-1} - R_r)/\kappa$, we form the effective incentive $\hat{\lambda}_r := \rho_r / \widehat{e_r^{\text{req}}(0)}$.

As a nonparametric check, we also estimate a local slope ratio at a common baseline $e = \Delta^{ibt}$ using finite differences on the fitted trajectories:

$$\hat{s}_i(\Delta^{ibt}) \approx \frac{\hat{v}_i(\Delta^{ibt} + h) - \hat{v}_i(\Delta^{ibt})}{h}, \quad \hat{\gamma}_r^{\text{slope}}(\Delta^{ibt}) := \frac{\hat{s}_r(\Delta^{ibt})}{\hat{s}_{r-1}(\Delta^{ibt})}.$$

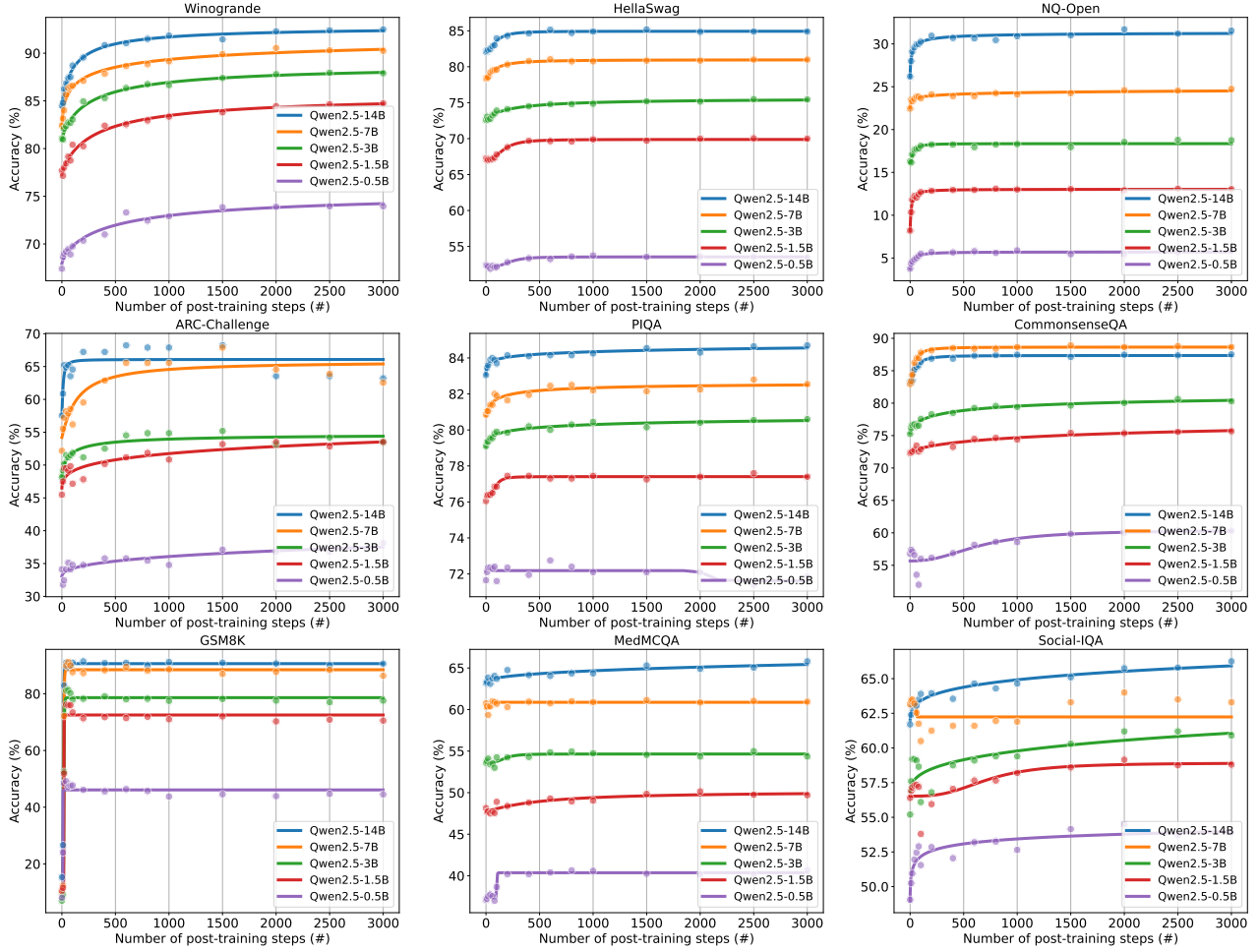


Figure 3. Continued post-training trajectories of *Qwen* models of different sizes on nine benchmarks. Here, we use model size as a proxy for the model’s latent capability θ . The x -axis denotes the number of post-training steps, reflecting post-training effort e . The y -axis denotes accuracy on the validation set, i.e., $v(\theta, e)$. For each model, we fit a curve following Equation (3).

E. Additional Empirical Results

Setting. We conduct our empirical study on nine benchmarks, Winogrande (Sakaguchi et al., 2019), HellaSwag (Zellers et al., 2019a), NQ-Open (Kwiatkowski et al., 2019), ARC-Challenge (Clark et al., 2018), PIQA (Bisk et al., 2019), CommonsenseQA (Talmor et al., 2019), Gsm8k (Cobbe et al., 2021), MedMCQA (Pal et al., 2022), and Social-IQA (Sap et al., 2019). Each model is trained with LoRA (Hu et al., 2021; Mangrulkar et al., 2022) (rank 8, $\alpha=32$) and AdamW optimizer (Loshchilov & Hutter, 2017) (weight decay 0.01 and learning rate $5e-5$). We use a batch size of 8, so each training step corresponds to eight training data points. For those benchmarks without a validation split, we randomly allocate 20% of the training data as the validation set.

Results. In Figure 3, we present the results of controlled post-training experiments for all nine benchmarks. The assumption 4.2 still largely holds in most benchmarks, despite a few anomalies. In addition, on benchmarks like HellaSwag, continued post-training cannot make a smaller model catch up to a larger one. This is not surprising, since model sizes (the proxy for θ) differ significantly in our controlled setting. As a result, there is little incentive to conduct strategic post-training in this scenario. We argue, however, that in real-world settings with models from diverse sources, θ will not differ so dramatically.

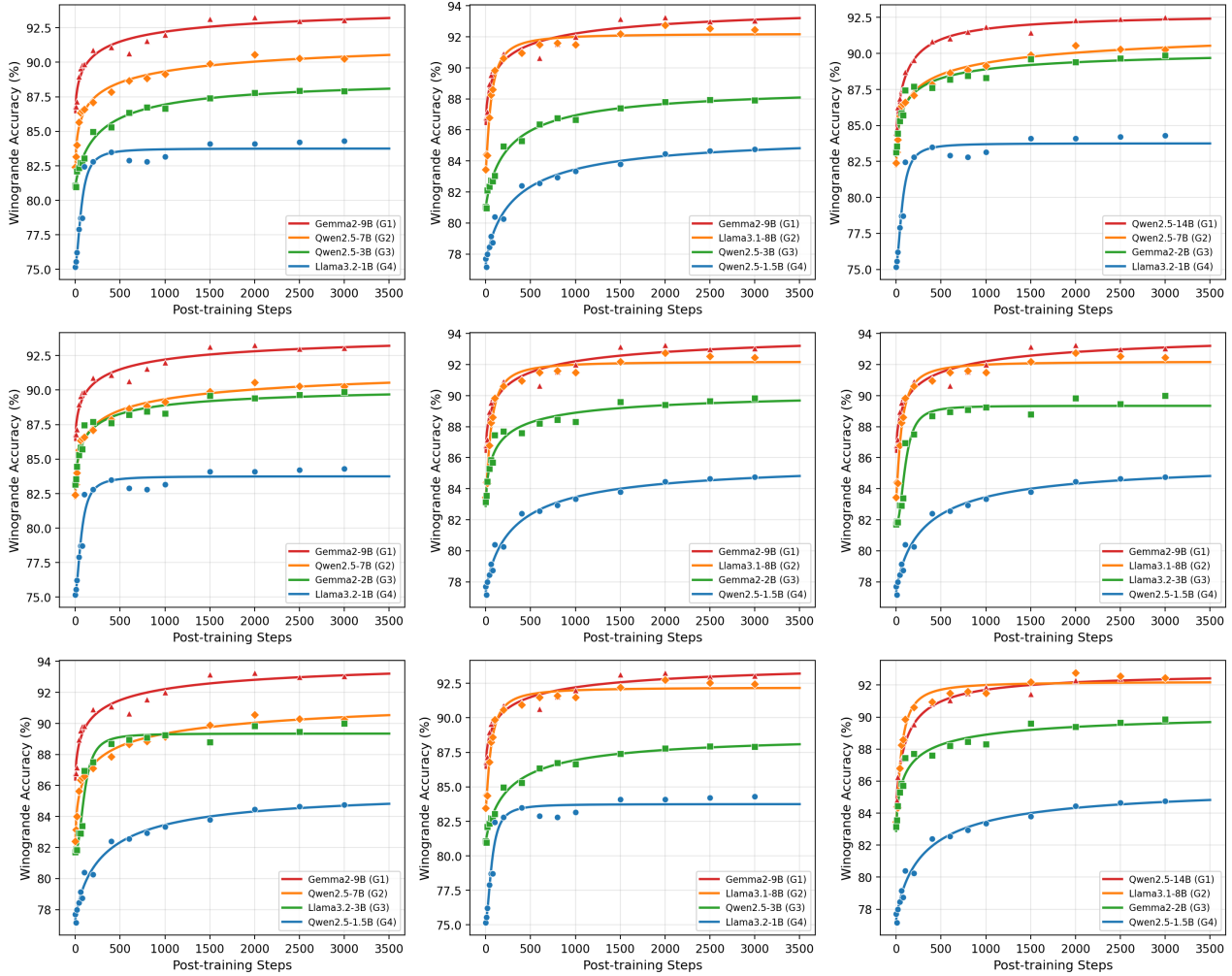


Figure 4. Continued post-training trajectories of models of different families on Winogrande benchmark. Here, we use PC1 from Zhang et al. (2025) as a proxy for the model’s latent capability θ . Based on PC1, we group 10 models into four tiers: G1 (Qwen2.5-14B, Gemma2-9B), G2 (Llama3.1-8B, Qwen2.5-7B), G3 (Qwen2.5-3B, Gemma2-2B, Llama3.2-3B), G4 (Qwen2.5-1.5B, Llama3.2-1B). We select one model per tier (ensuring all three families appear) and apply the same SFT recipe on Winogrande across all 9 valid combinations. The x -axis denotes the number of post-training steps, reflecting post-training effort e . The y -axis denotes accuracy on the validation set, i.e., $v(\theta, e)$. For each model, we fit a curve following Equation (3).

E.1. Extension to Multi-Family Models

We further validate the Assumption 4.2 across families. Because latent capability θ is unobserved, we use PC1 of the score matrix across 24 benchmarks by Zhang et al. (2025) as a proxy for latent capability. PC1 captures the dominant variation axis, so models scoring consistently higher across diverse tasks have higher PC1 values, independent of family. Based on PC1, we group 10 models into four tiers: G1 (Qwen2.5-14B, Gemma2-9B), G2 (Llama3.1-8B, Qwen2.5-7B), G3 (Qwen2.5-3B, Gemma2-2B, Llama3.2-3B), G4 (Qwen2.5-1.5B, Llama3.2-1B). We select one model per tier (ensuring all three families appear) and apply the same SFT recipe on Winogrande across all 9 valid combinations.

The results are shown in Figure 4. Curves may cross early because models from different families have undergone different prior post-training (instruction tuning, RLHF, etc.) that we cannot offset. Despite this, C2 holds universally as all models exhibit clear diminishing returns. Besides, the final ordering consistently matches $G4 < G3 < G2 < G1$ across all 9 combinations, meaning higher-capability models benefit more from training and overtake those with a head start, which is exactly what C3 implies. This also directly demonstrates why TbT works: after sufficient uniform fine-tuning, the true capability ranking emerges regardless of prior effort.

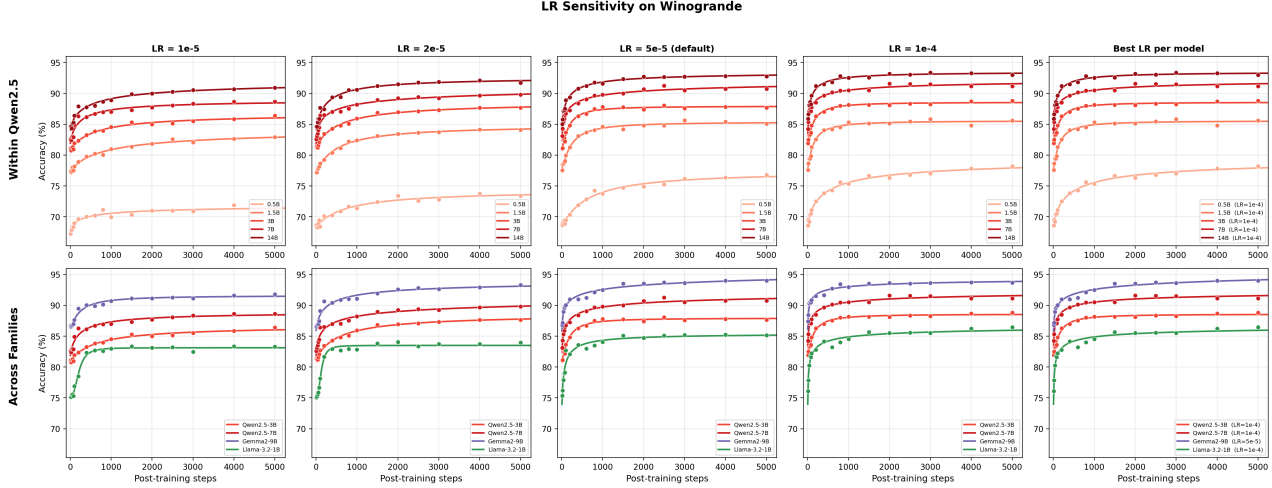


Figure 5. LR sweep on Winogrande for within-family (Qwen2.5: 0.5B, 1.5B, 3B, 7B, 14B; top) and cross-family (Gemma2-9B, Qwen2.5-7B, Qwen2.5-3B, Llama3.2-1B; bottom) settings. The x -axis is the number of post-training steps (training effort e); the y -axis is validation accuracy $v(\theta, e)$. For each model and LR, we fit a curve following Equation (3). The rightmost column uses, for each model, the LR maximising validation accuracy at step 5000.

E.2. Sweeping Learning Rate

We sweep $\text{LR} \in \{1\text{e-}5, 2\text{e-}5, 5\text{e-}5, 1\text{e-}4\}$ in both within-family and cross-family settings on Winogrande (Figure 5). The within-family row preserves the size order $14\text{B} > 7\text{B} > 3\text{B} > 1.5\text{B} > 0.5\text{B}$ at every LR. In the cross-family row, the step-5000 ranking Gemma2-9B > Qwen2.5-7B > Qwen2.5-3B > Llama3.2-1B, corresponding to the capability tiers $\text{G1} > \text{G2} > \text{G3} > \text{G4}$, is preserved at every LR. We also select the best LR for each model individually based on step-5000 validation accuracy (rightmost column in each figure). The ranking remains unchanged, confirming that the rankings induced by continued post-training are robust to the choice of learning rate.